Zoning and the Dynamics of Urban Redevelopment

Vincent Rollet

MIT

Many large cities fail to make space for its new residents

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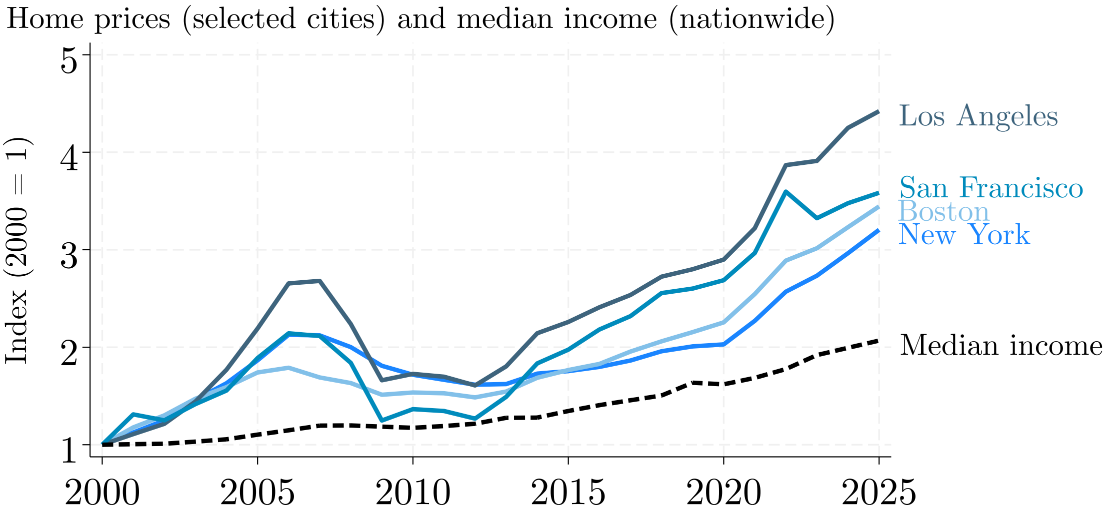

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Zoning: Regulations restricting characteristics of new buildings.

- **Density limits** (e.g., height limits).
- **Use limits** (e.g., residential only).

THE WALL STREET JOURNAL.

Only Zoning Reform Can Solve America's Housing Crisis

Los Angeles Times

L.A. can't become an affordable city by protecting single-family zoning

FT FINANCIAL TIMES

Planning rules are driving the global housing crisis

The Boston Globe

Do homes really need 2-acre lots? Spoiler alert: No.

Technological constraints matter too

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- Change comes from **redevelopment**.
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- Increasingly how cities grow (Frolking et al., 2024).
- Requires paying large **fixed costs**.

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  - → **Demand**: Spatial equilibrium model predicts how development affects prices.

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Policy application: What would be the effects of relaxing zoning?

- **Quasi-experimental evidence** from recent zoning changes.
- Model accurately predicts reduced-form effects.
- Use model to predict the **effects of relaxing zoning** on **construction** and **prices**.

Findings

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- But **adjustment** to policy change is **slow**, **take-up of upzoning is limited**.
  - → Only 1/3 of long-term gains in first decade.
  - → Allowing 5 new units yields only 1 built over 40 years.
  - → Mechanism: Large **fixed costs** that rise steeply with height.
  - → Implication: Deregulation most effective where **prices are high** and **density is low**.

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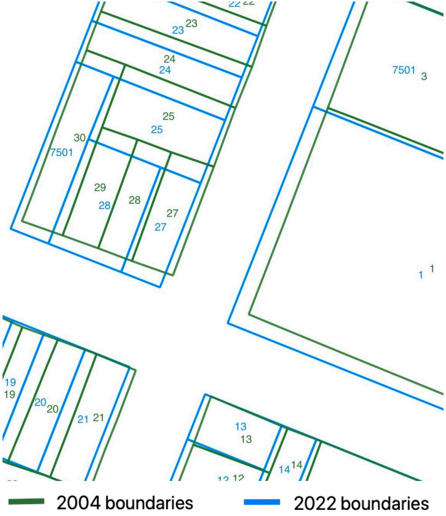
- Affordability **gains geographically spread out** due to migration.

Data and context

Land use

- I use **cadastre maps** to:
  - → Partition NYC in time-consistent parcels.
  - → Link land parcels to buildings over time.



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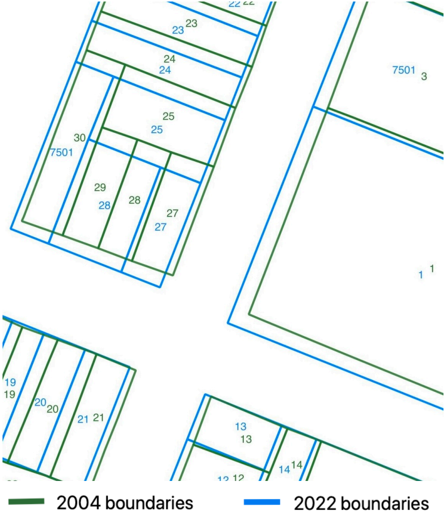
- Track universe of parcels over 2004–2022.
  - → 833,000 parcels in total.



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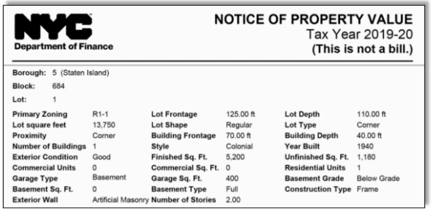
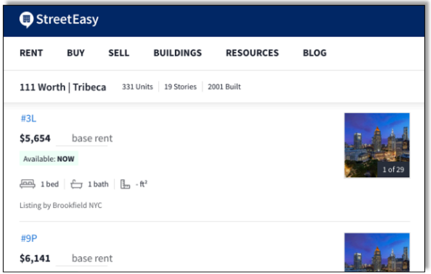
- **Building characteristics**:
  - → Property tax records (scraping, FOIA requests).
  - → StreetEasy (scraped data on 1.6 million units).


NYC
Department of Finance

NOTICE OF PROPERTY VALUE
Tax Year 2019-20
(This is not a bill.)

Borough: 5 (Staten Island)
Block: 684
Lot: 1

Primary Zoning	R1-1	Lot Frontage	125.00 ft	Lot Depth	110.00 ft
Lot square feet	13,750	Lot Shape	Regular	Lot Type	Corner
Proximity	Corner	Building Frontage	70.00 ft	Building Depth	40.00 ft
Number of Buildings	1	Style	Colonial	Year Built	1940
Exterior Condition	Good	Finished Sq. Ft.	5,200	Unfinished Sq. Ft.	1,180
Commercial Units	0	Commercial Sq. Ft.	0	Residential Units	1
Garage Type	Basement	Garage Sq. Ft.	400	Basement Grade	Below Grade
Basement Sq. Ft.	0	Basement Type	Full	Construction Type	Frame
Exterior Wall	Artificial Masonry	Number of Stories	2.00		



Redevelopment

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Redevelopment

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  → Buildings constructed since 2004.
- **Building permits**
- **Certificates of Occupancy**
  → Construction ends.
  → Scraped and digitized 250,000 documents.

Construction timelines for 22,000 redevelopment events

Redevelopment duration | Quantity changes | Quantity changes, by use

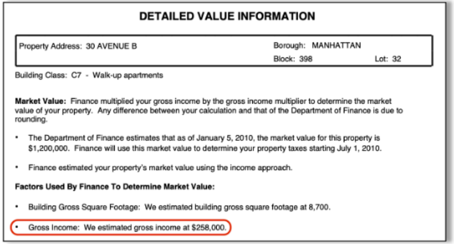
Prices

- **Sales prices** from real estate transactions over 2003-2022 (1.2 million observations).

  $\rightarrow$ From NYC Department of Finance.

Prices

- **Sales prices** from real estate transactions over 2003-2022 (1.2 million observations).
  - → From NYC Department of Finance.
- Yearly **rent** measures for 73,000 buildings.
  - → Scraped 600,000 tax documents.



DETAILED VALUE INFORMATION

Property Address: 30 AVENUE B   Borough: MANHATTAN
Block: 398   Lot: 32

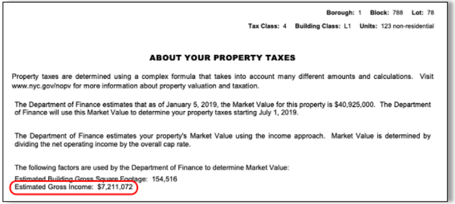
Building Class: C7 - Walk-up apartments

Market Value: Finance multiplied your gross income by the gross income multiplier to determine the market value of your property. Any difference between your calculation and that of the Department of Finance is due to rounding.

- The Department of Finance estimates that as of January 5, 2010, the market value for this property is $1,200,000. Finance will use this market value to determine your property taxes starting July 1, 2010.
- Finance estimated your property's market value using the income approach.

Factors Used By Finance To Determine Market Value:

- Building Gross Square Footage: We estimated building gross square footage at 8,700.
- Gross Income: We estimated gross income at $258,000.



Borough: 1 **Block:** 788 **Lot:** 78
Tax Class: 4 **Building Class:** L1 **Units:** 123 non-residential

ABOUT YOUR PROPERTY TAXES

Property taxes are determined using a complex formula that takes into account many different amounts and calculations. Visit www.nyc.gov/nopv for more information about property valuation and taxation.

The Department of Finance estimates that as of January 5, 2019, the Market Value for this property is $40,925,000. The Department of Finance will use this Market Value to determine your property taxes starting July 1, 2019.

The Department of Finance estimates your property's Market Value using the income approach. Market Value is determined by dividing the net operating income by the overall cap rate.

The following factors are used by the Department of Finance to determine Market Value:
Estimated Building Gross Square Footage: 154,516
Estimated Gross Income: $7,211,072

Zoning

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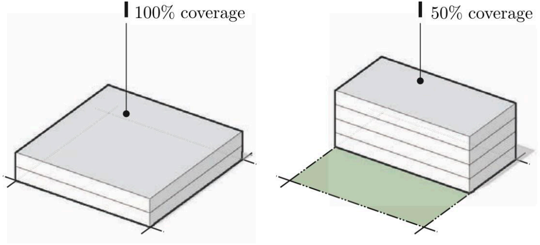
- Zoning limits the **size** and **use** (residential vs. commercial) of new buildings.
- Main zoning instrument in NYC: limits on the **Floor Area Ratio** (FAR).

$$\textbf{FAR} = \frac{\text{sq. ft of floorspace}}{\text{sq. ft of land}}$$

Buildings with a FAR of 2



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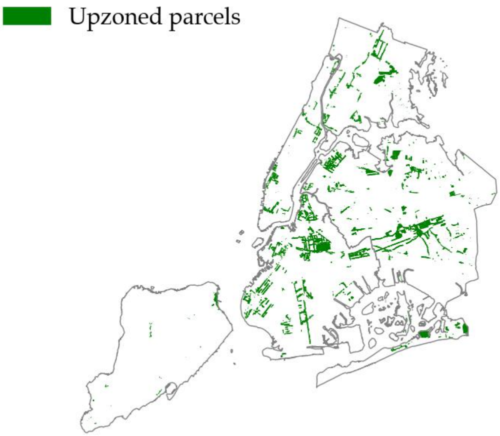
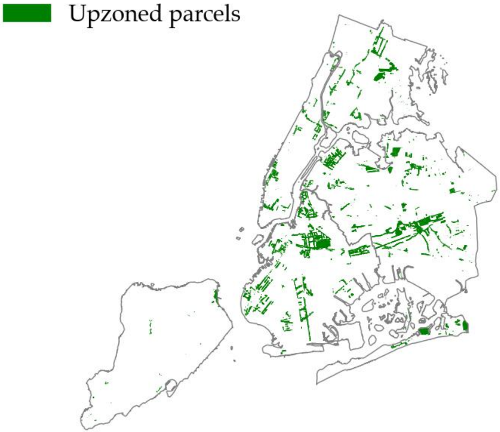


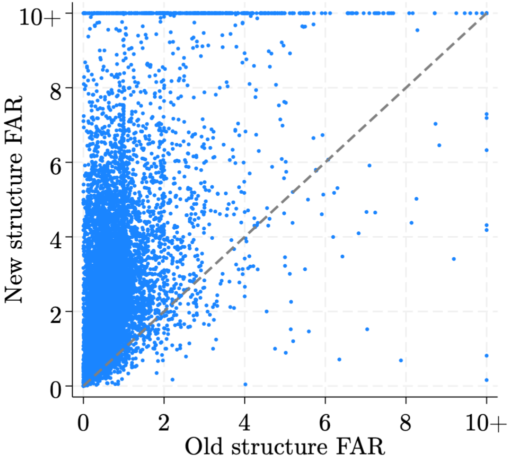
- Zoning ordinance adopted in 1961.
  → Strong persistence over time.
- Several areas upzoned in past decades.
  → Focus on large planner-initiated upzonings.
- Many veto players.
  → Exact changes and timing unclear when upzoning discussions begin.



Empirical evidence

Redevelopment means densification

- New buildings 3.4 times larger than the ones they replace (on average).

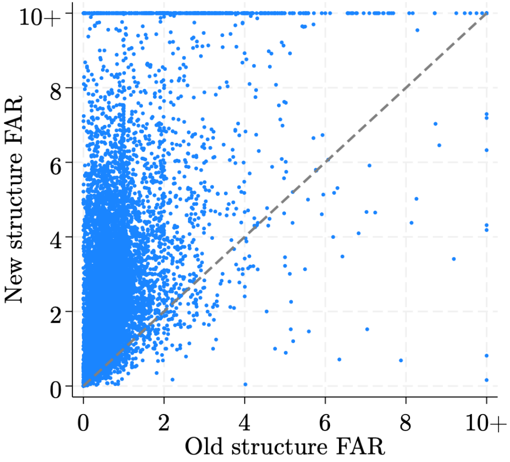


Change in number of units

Redevelopment means densification

- New buildings 3.4 times larger than the ones they replace (on average).

- Suggests large redevelopment **fixed costs**.



Change in number of units

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- Compare parcels upzoned earlier vs. later.
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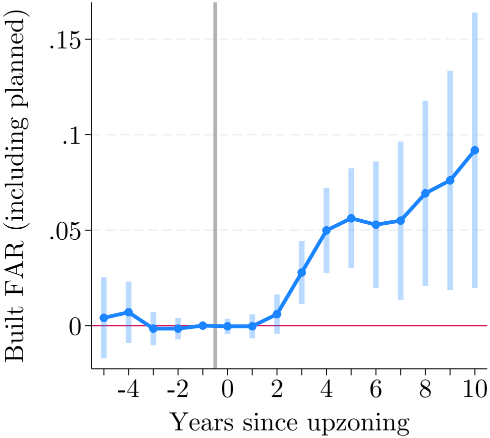
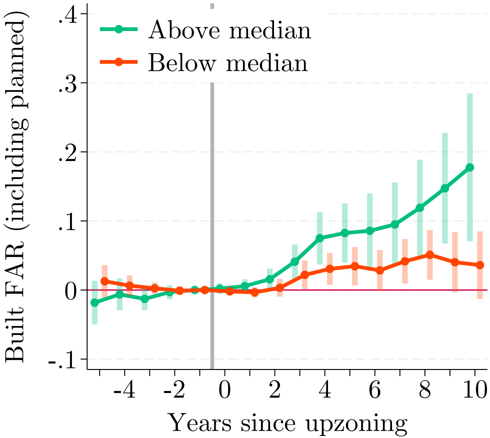
- ≈ 10% of increase used after a decade.



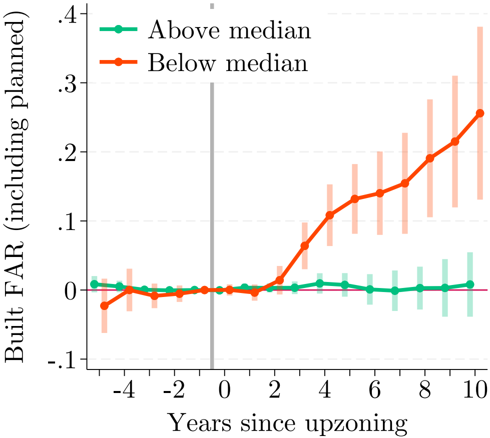
The effects of upzoning vary widely

Effects concentrate in high-price neighborhoods and underbuilt parcels

Heterogeneity by baseline neighborhood floorspace prices



Heterogeneity by baseline built FAR (within neighborhood)



A dynamic model of urban change

Supply of floorspace

Dynamic behavior of developers given prices and regulation

Developer choices

Each year, atomistic developers make independent redevelopment decisions for each parcel.

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$\downarrow$

Build?

Extensive margin

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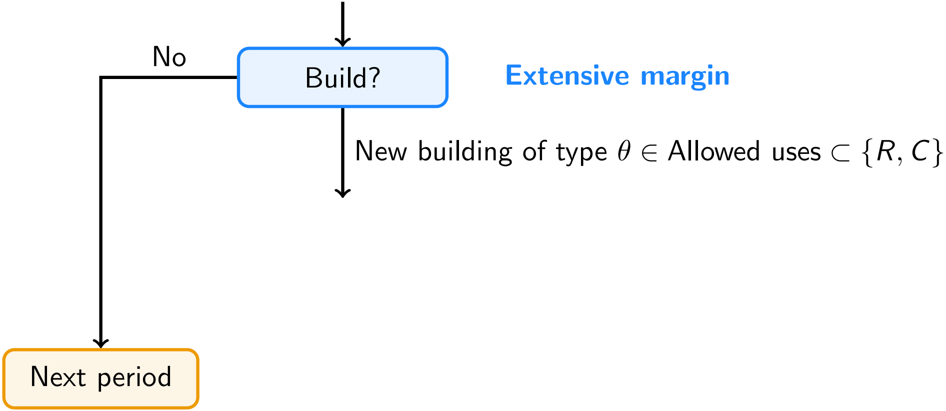

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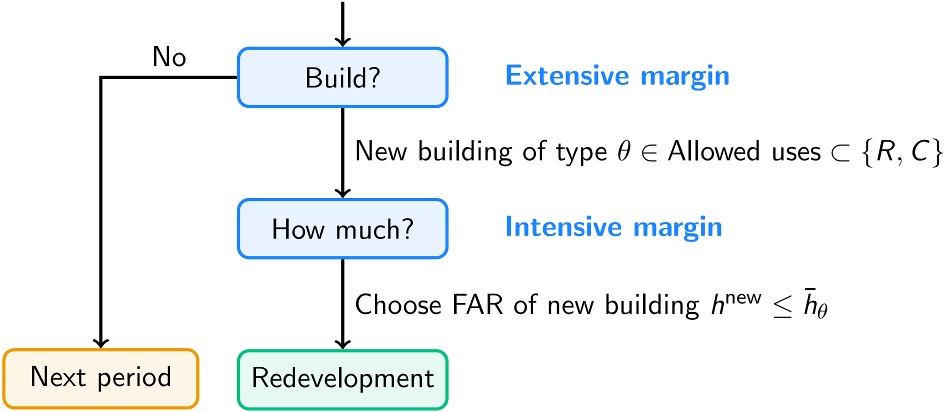

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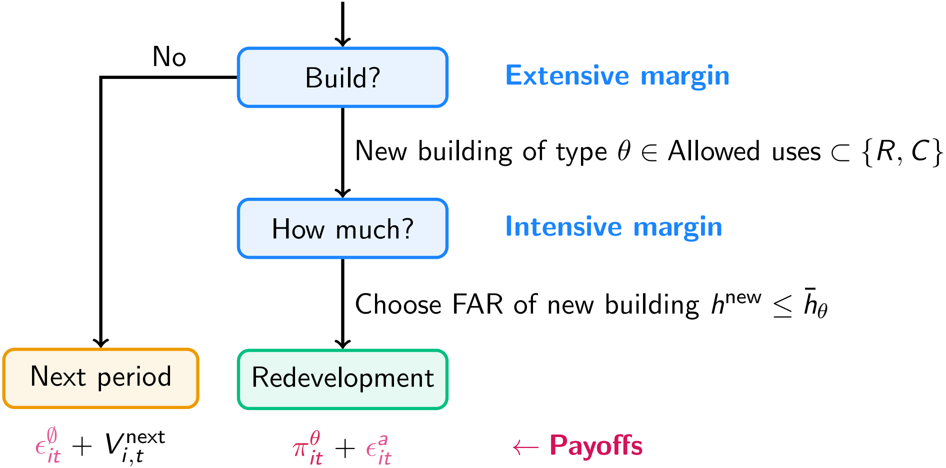

Developer profit function

$$\pi_{it}^{\theta} = \max_{h^{\text{new}} \leq \bar{h}_{\theta it}} \left[\underbrace{P_{it}^{\text{new}}(h^{\text{new}}) - P_{it}^{\text{old}}}_{\text{Change in property value}} - \underbrace{[VC_{it}(h_{it}^{\text{new}}) + FC_{it}]}_{\text{Cost of redevelopment}} \right]$$

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- Variable costs (VC): construction costs, increase with size of new building.
- Fixed costs (FC): eviction, demolition, permitting, etc.
  → Vary by building (e.g., larger for bigger buildings in dense neighborhoods).

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Estimation: Prices

- Value of a building:

$$P_{it} = \underbrace{\rho_{nt}^{\theta}}_{\text{Location premium}} \times \underbrace{Q(\mathbf{x}_{it})}_{\text{Building quality}} \times \underbrace{h_{it}}_{\text{Building size}}$$

→ Estimated through a hedonic regression on the sales data.
→ Focus on buildings at/above FAR limit (option value of redevelopment ≈ 0).

Regression equation | Hedonic parameter estimates

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- Characteristics in $\mathbf{x}_{it}$:

→ Building age
→ Building grade
→ Building material
→ Building type
→ (log) unit size
→ (log) FAR
→ Landmark status
→ Rent-stabilized

Regression equation
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- **Parametric assumption**: Cobb-Douglas production of floorspace with capital and land.

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- Recover parameters through maximum likelihood estimation.
  → Developers choose FAR $h \leq \bar{h}$ to max profits.

Identifying variation | Parameter estimates | Estimated construction costs | Skyscraper construction costs

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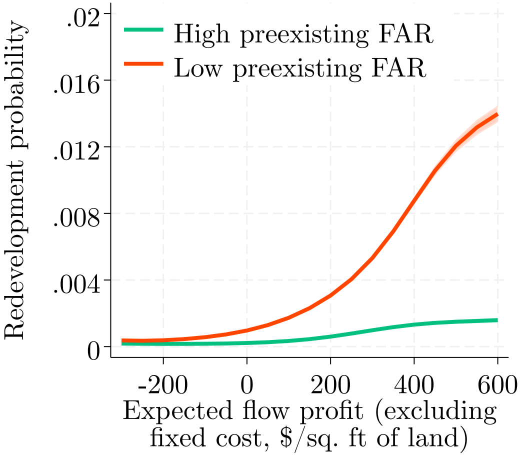

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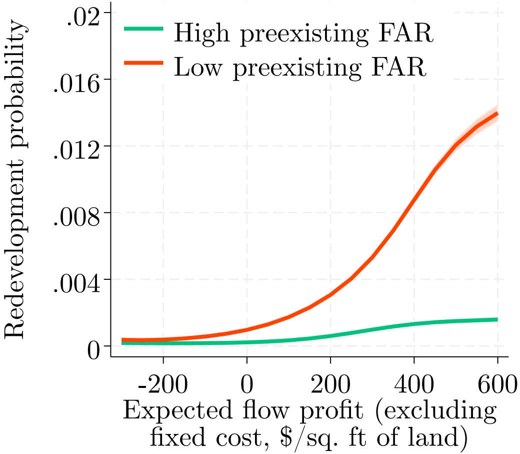
  - → Increase with size of old building.
  - → Increase with neighborhood density.
  - → Larger in historic districts.



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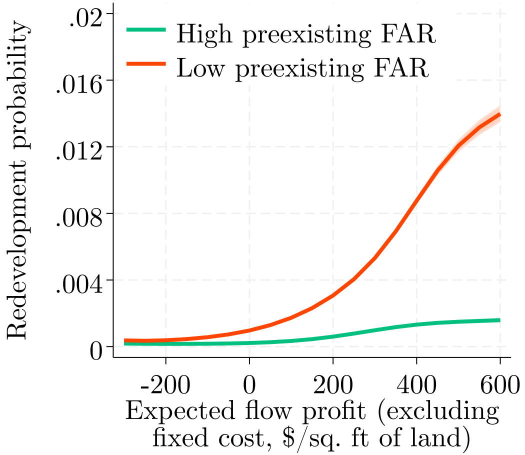
  - → Larger in historic districts.
- Estimation using **full-solution approach** (extending Rust, 1987).
  - → Inner loop: Value function estimation.
  - → Outer loop: Maximum likelihood.



Demand for floorspace

Evolution of prices given developers' choices

Demand model: Overview More

- Workers consume residential floorspace, choose where to live/work.
- Firms produce using commercial floorspace and labor.
- **Key extensions**: Heterogeneous types and non-homothetic preferences for housing.
  → Essential to rationalize consumed quantities of housing.
- Additional ingredients:
  → Migration
  → Congestion
  → Agglomeration externalities (internally estimated). More

Model validation

Predicted effect of upzonings

- Model estimation separate from reduced-form.
  $\rightarrow$ 90% of redevelopment on non-upzoned parcels.

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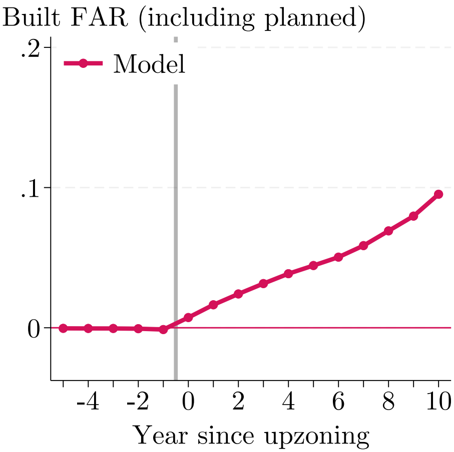
  3. Build model-implied effect of upzoning.



In-sample fit | Residential vs. commercial | Fit by borough | Out-of-sample fit | Parcel-level | Aggregates

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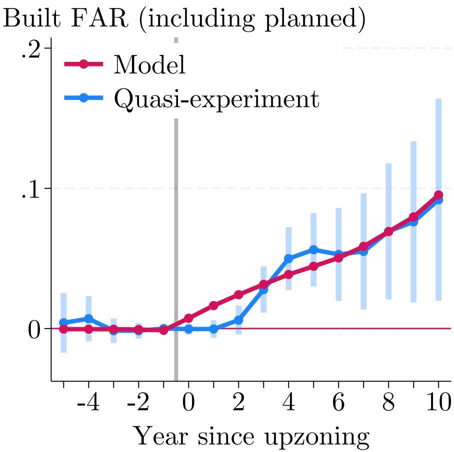
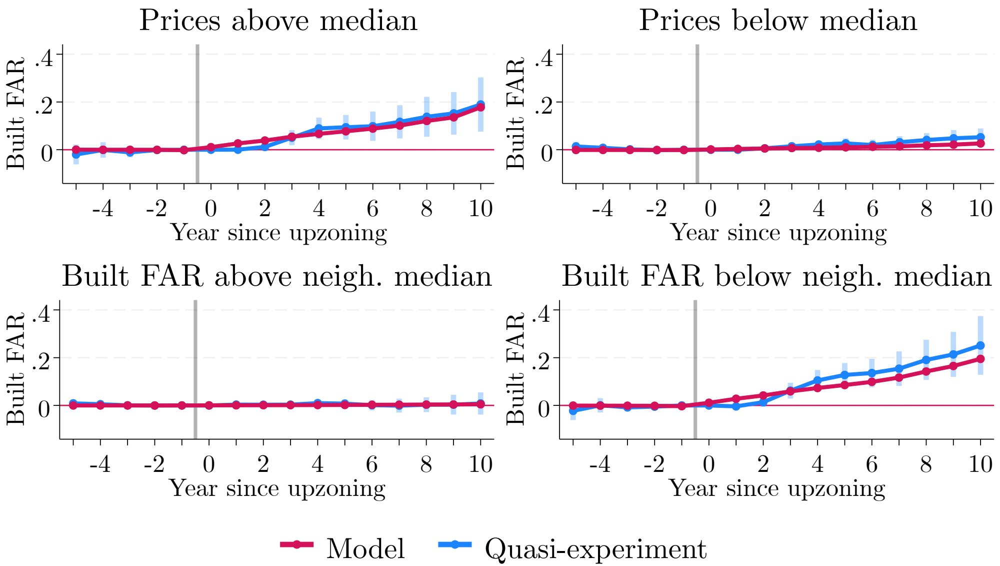
- **Validation exercise**:
  1. Estimate model on non-upzoned parcels.
  2. Predict evolution of upzoned parcels with/without upzoning.
  3. Build model-implied effect of upzoning.
  4. Compare with quasi-experimental estimates.



In-sample fit | Residential vs. commercial | Fit by borough | Out-of-sample fit | Parcel-level | Aggregates

Model validation: Predicted effect of upzonings



Implications for zoning policy

Main counterfactuals

Simulate the evolution of the city until 2060, keeping fundamentals at their 2019 level.

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Upzoning map | Protected parcels | Flood zones

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---

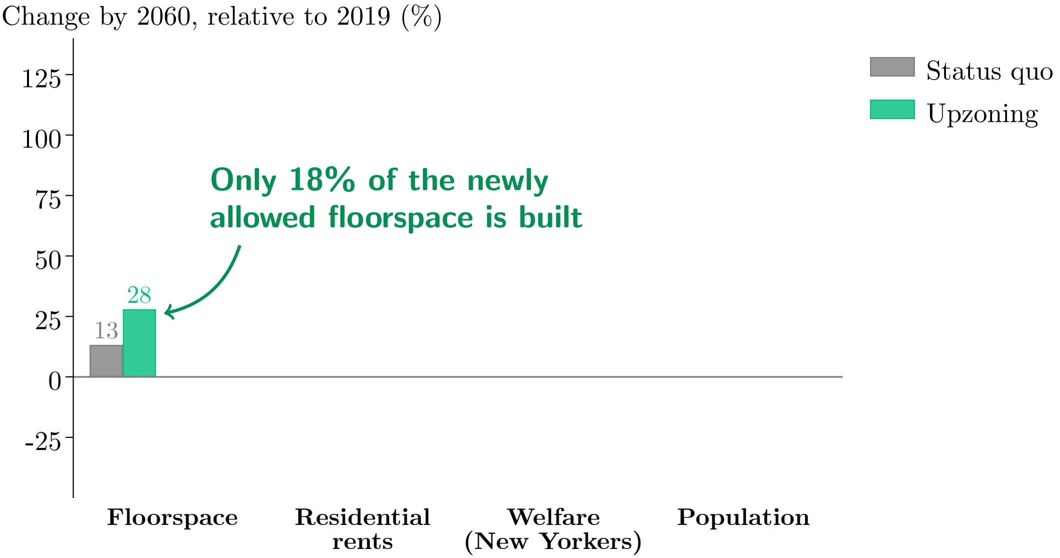
4. **Frictionless benchmark** (no zoning, no adjustment costs, price = marginal cost).

Upzoning map | Protected parcels | Flood zones

To what extent does zoning constrain NYC's growth?

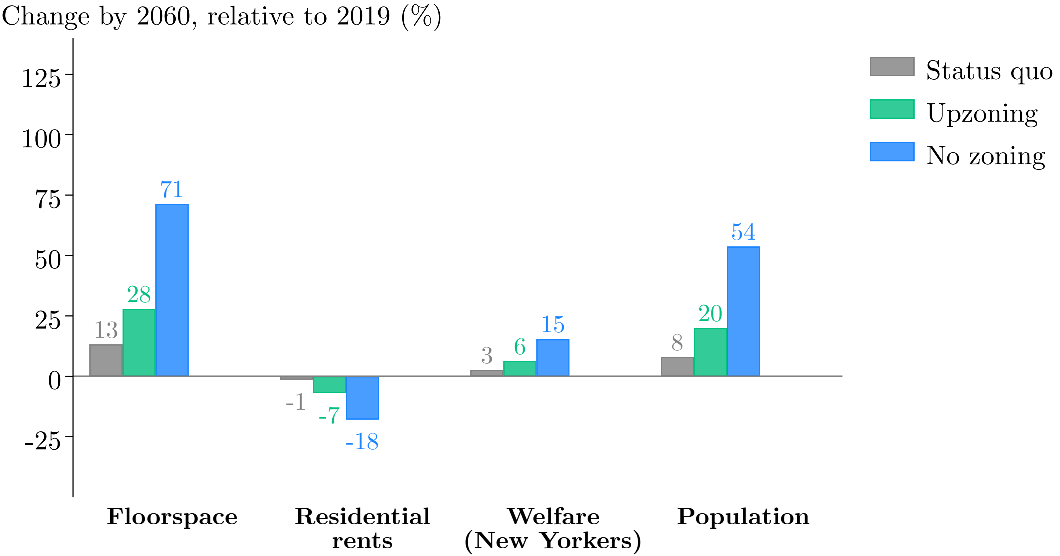
Under current zoning, NYC continues to grow slowly

Upzoning increases floorspace by 15pp over 40 years

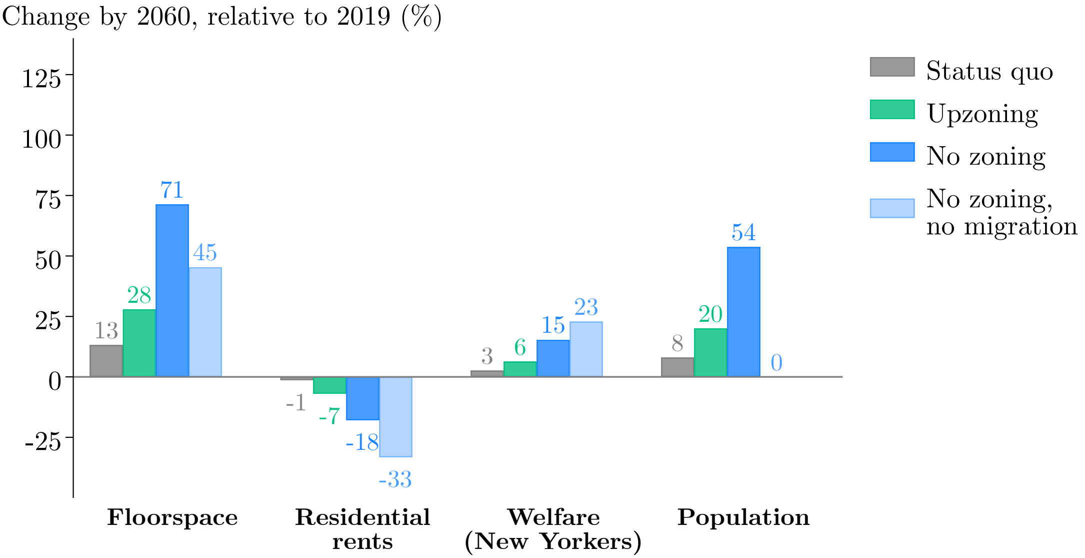


Completely removing zoning quintuples NYC's growth rate

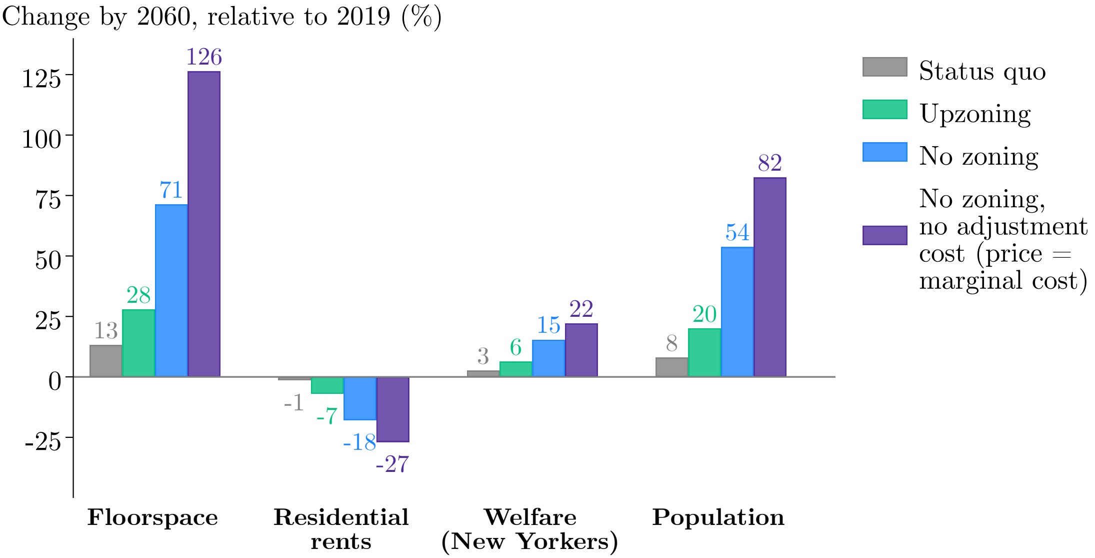
Rent decreases are moderate



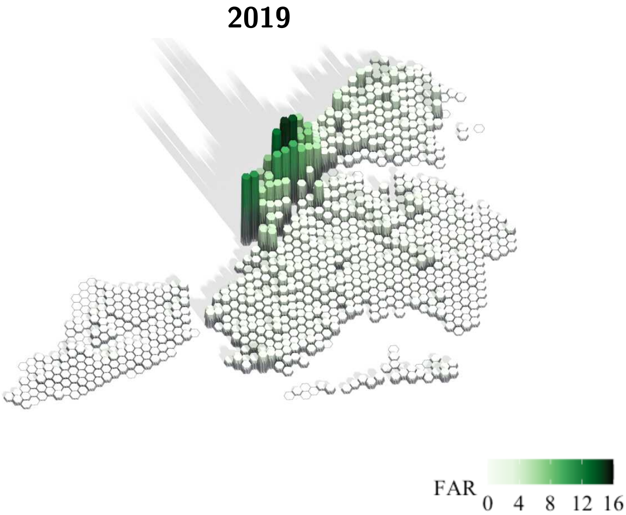
Rent decreases are moderated by migration



A simpler model greatly overstates effects of removing zoning



Technological constraints matter for credible counterfactuals



More

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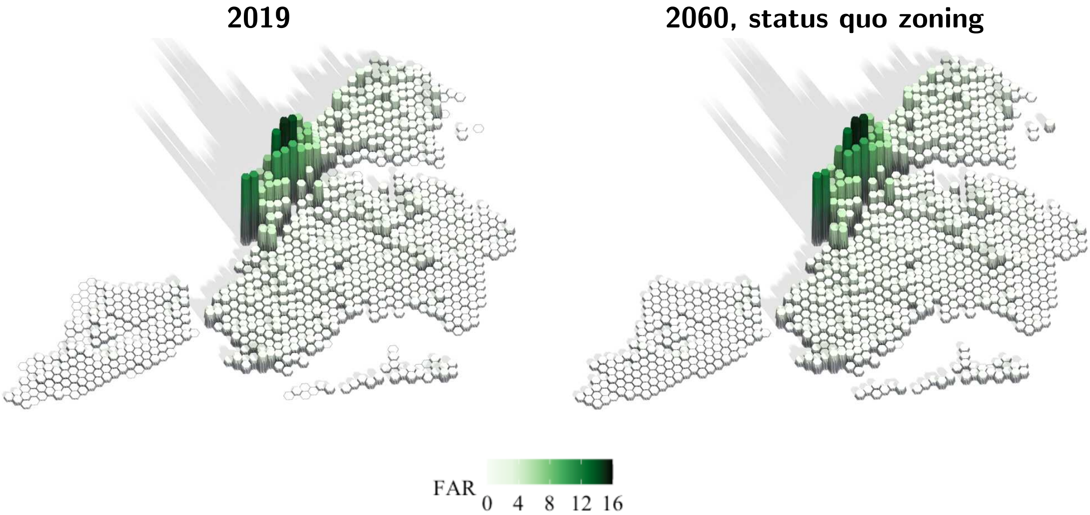

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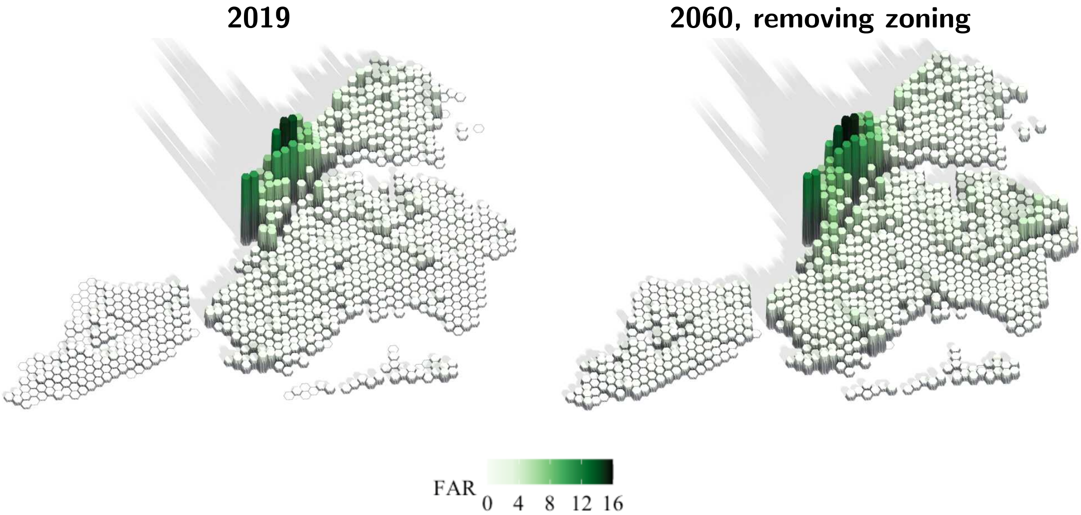

More

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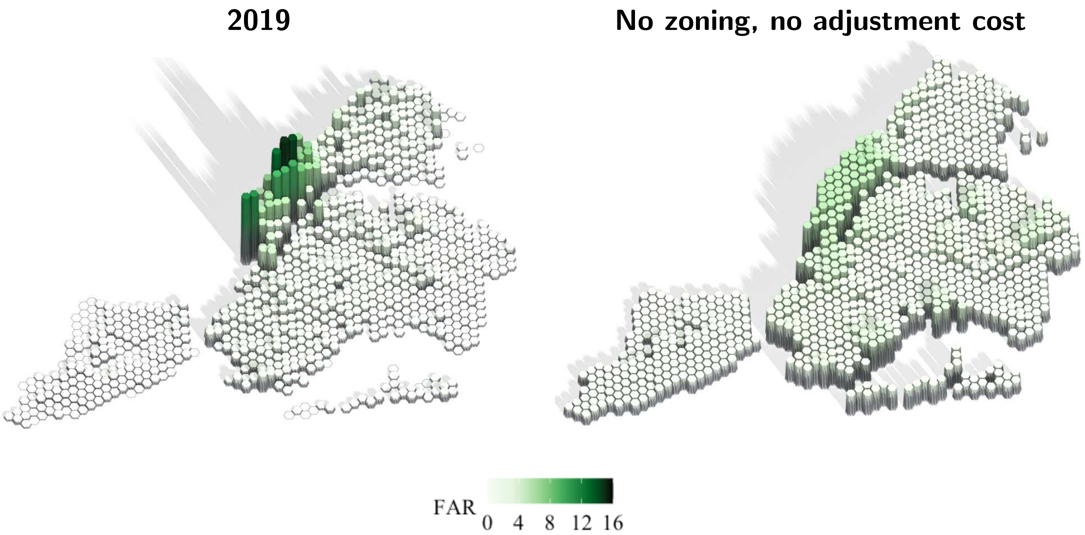
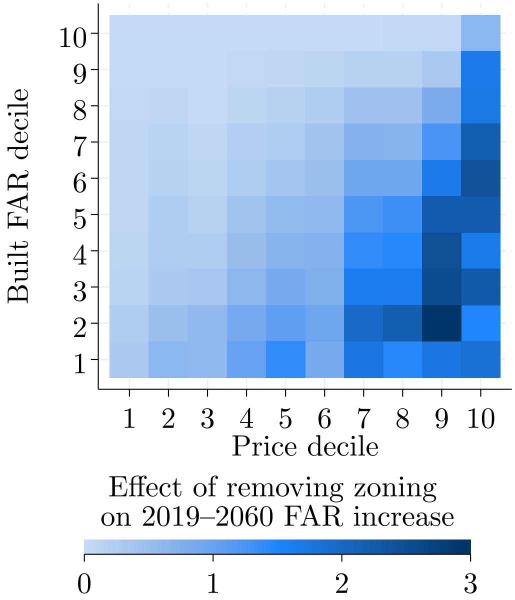


More

Where is zoning a constraint?

Where does relaxing zoning lead to increased supply?



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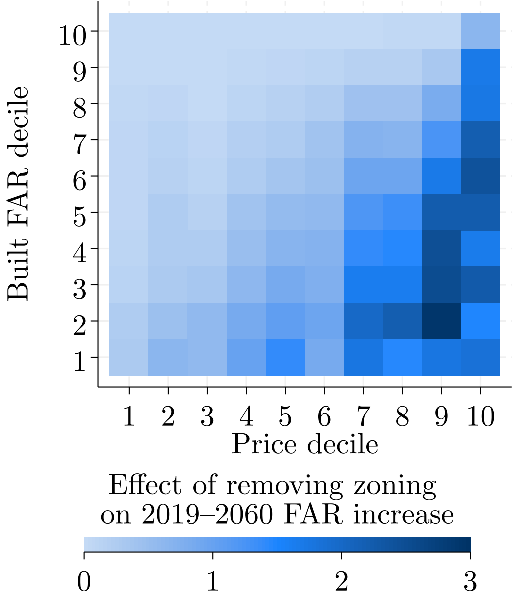

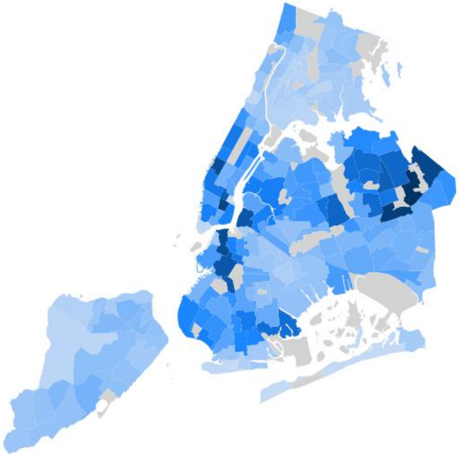

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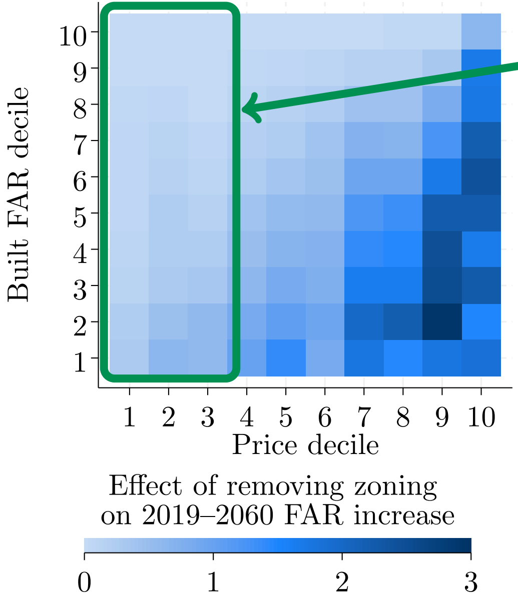
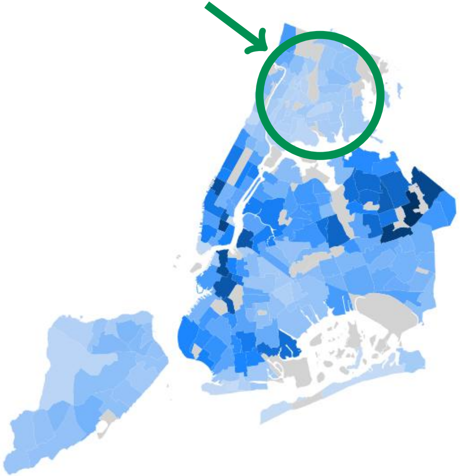


Zoning tends not to be binding when floorspace prices are low.



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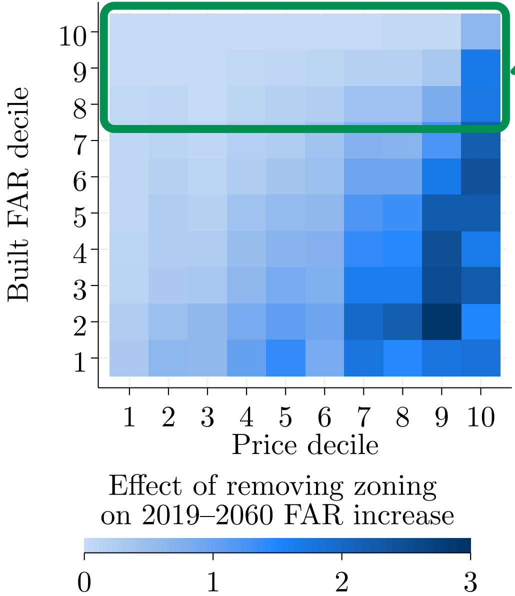
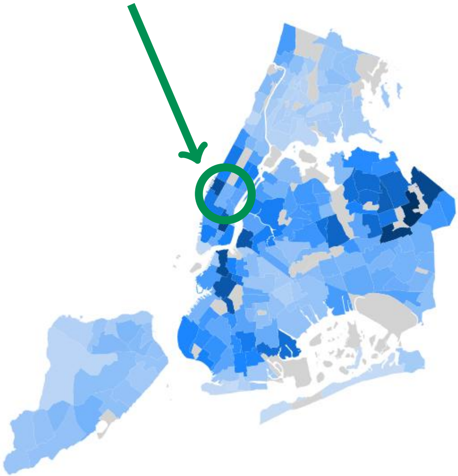


Removing zoning has little effect on parcels with tall buildings.



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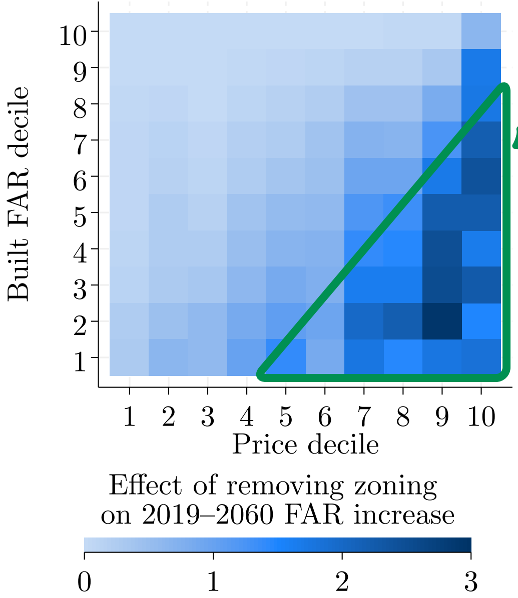
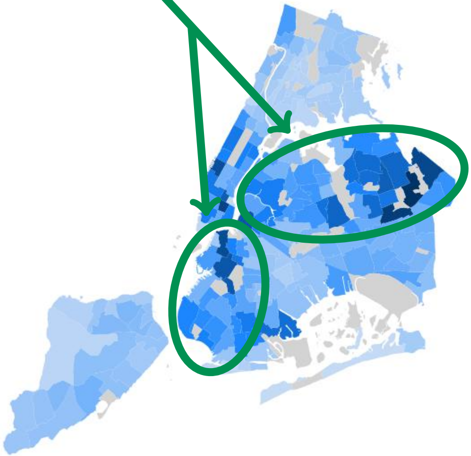


Upzoning is effective when prices are high enough and density is low enough.



Conclusion

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  - → Even at a 40-year horizon, **take-up of upzoning is limited**.
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  - → Upzoning works best when **prices are high/density is low**.
- Welfare gains **take time to materialize** and are **geographically spread out**.
  - → Locally, effects of upzoning might look disappointing despite **large national welfare gains**.

Appendix

Demand for floorspace ◂ Back

- Neighborhoods endowed with commercial and residential floorspace.

- **Key extension**: Non-homothetic preferences and strong worker heterogeneity.
  → Essential to rationalize consumed quantities of housing.

Demand for floorspace

Back

- Neighborhoods endowed with commercial and residential floorspace.

- Workers consume residential floorspace, choose home and work locations (i, j):

$$\max_{i,j,h,c} U = \underbrace{B_i}_{\text{amenities}} \cdot \underbrace{(H - \underline{H}_i)^{\beta}}_{\text{housing}} \cdot \underbrace{c^{1-\beta}}_{\text{other cons.}} \cdot \underbrace{d_{ij}^{-1}}_{\text{commuting}} \cdot \underbrace{z_i^H z_j^W}_{\substack{\text{idiosyncratic}\\ \text{preferences}}}$$

$$\text{s.t. } c + R_i H \leq \text{Income}$$

→ Labor income $= s(\vartheta) w_j$, skill $s(\vartheta)$ lognormally distributed.
→ Income from rented floorspace redistributed proportionally to labor income.

- **Key extension**: Non-homothetic preferences and strong worker heterogeneity.
  → Essential to rationalize consumed quantities of housing.

Amenities and productivities ◂ Back

- Firms consume commercial floorspace:

$$Y_j = A_j \cdot \underbrace{H_{Fj}^{\alpha_j}}_{\substack{\text{Commercial} \\ \text{floorspace}}} \cdot \underbrace{L_{Fj}^{1-\alpha_j}}_{\text{Labor}}$$

Amenities and productivities

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- Firms consume commercial floorspace:

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- Amenities (B) and productivities (A) vary with density of residents ($\tilde{L}_R$) and jobs ($\tilde{L}_F$).

$$B_i = \bar{B}_i \tilde{L}_{Ri}^{\gamma_{RR}} \tilde{L}_{Fi}^{\gamma_{CR}} \qquad A_j = \bar{A}_j \tilde{L}_{Rj}^{\gamma_{RC}} \tilde{L}_{Fj}^{\gamma_{CC}}$$

→ γ are the agglomeration elasticities.

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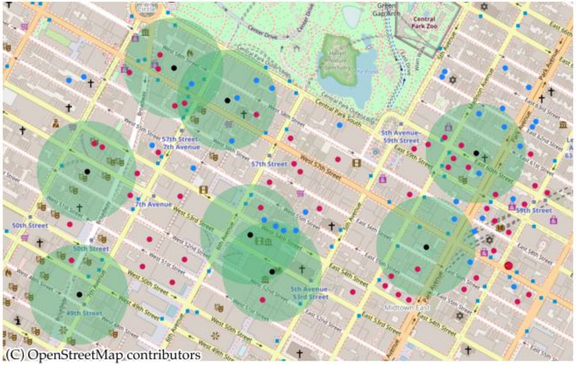
- The calibrated model matches untargeted moments well.

Price effects of new construction

◂ Back

- To calibrate agglomeration externalities γ, measure local demand elasticities:
  - → Draw 500-ft disks around large new construction events.
  - → Compare the evolution of rents in disks treated earlier vs. later.

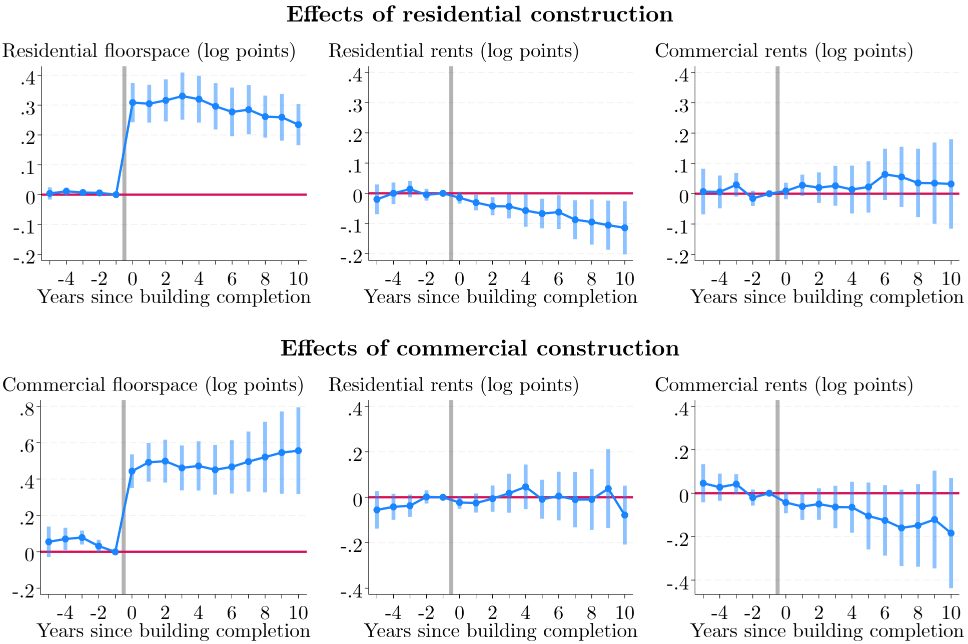


Events (new residential buildings)

- New buildings
- 500 ft buffers

Buildings with available rent data

- Residential
- Commercial


Effects of residential construction
Residential floorspace (log points)
Residential rents (log points)
Commercial rents (log points)
Years since building completion
Effects of commercial construction
Commercial floorspace (log points)
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Price effects of new construction

Back

- I calibrate agglomeration parameters through **indirect inference** to match reduced-form elasticities.

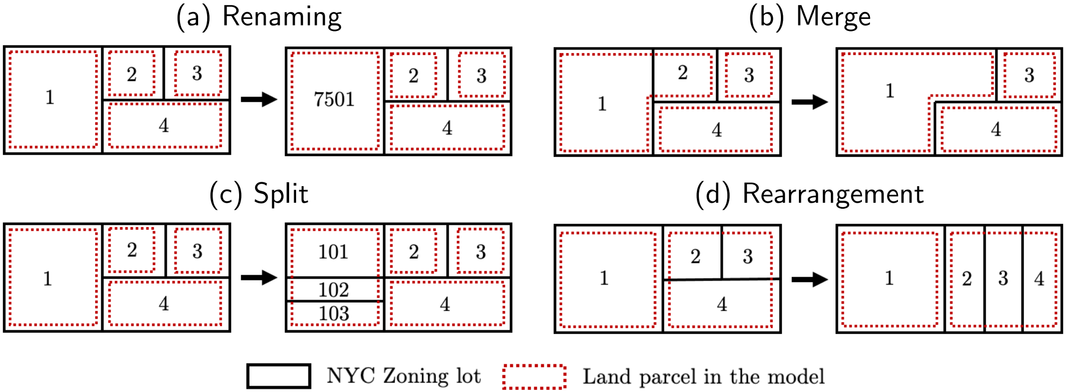
Price effects of new construction

Back

- I calibrate agglomeration parameters through **indirect inference** to match reduced-form elasticities.

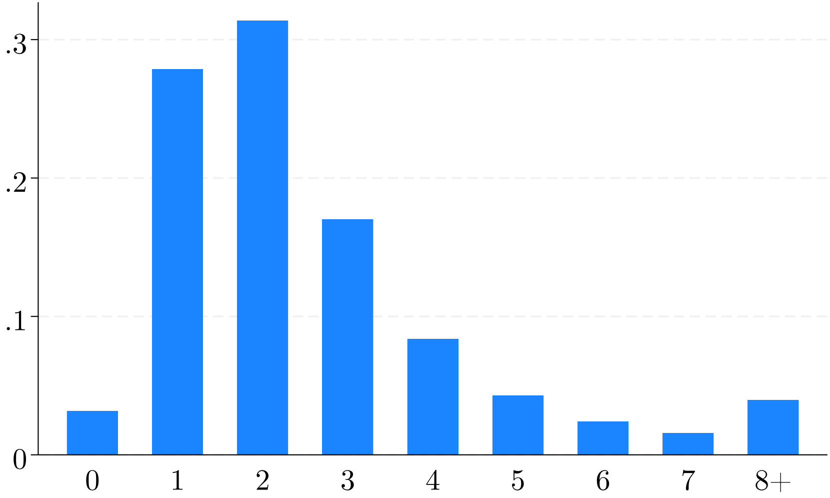
- I find agglomeration externalities in line with existing estimates in the literature.
  - → Effect of residents on other residents: $\gamma_{RR} = 0.11$.
  - → Effect of firms on other firms: $\gamma_{CC} = 0.07$.
  - → Corresponding estimates in Ahlfeldt et al. (2015): 0.16 and 0.07.

Boundary changes (21,700 detected over 2004-2022)



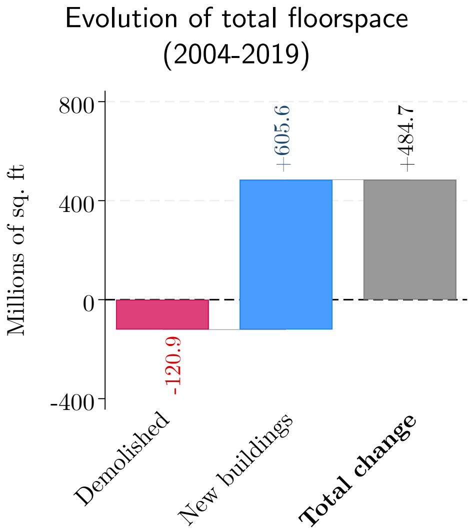
Back

Redevelopment duration (in years)



Back

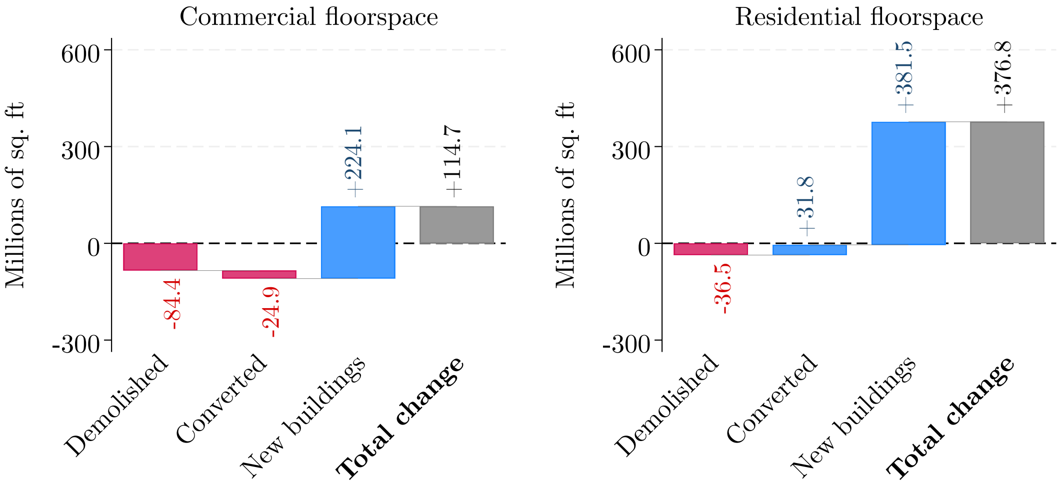
Mature cities grow by redeveloping old structures



- Since 2004, floorspace in NYC has grown at a rate of ∼0.6% per year.

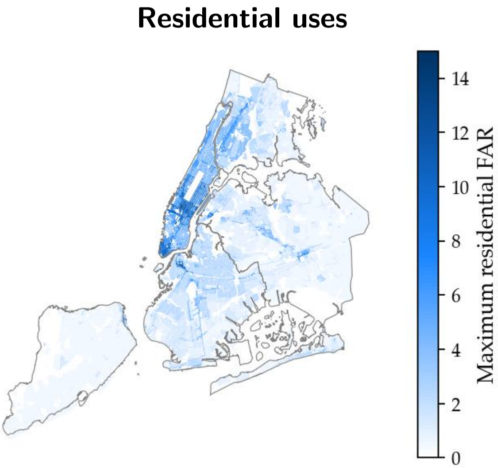
Back

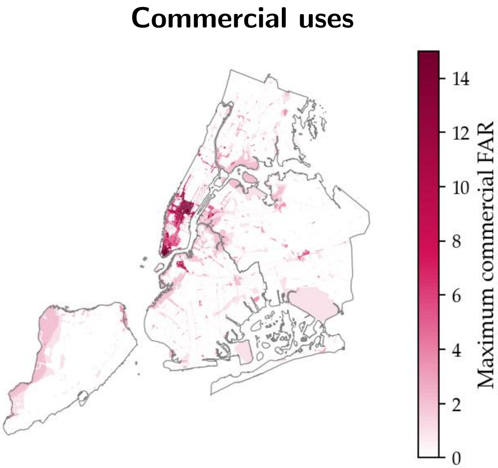
Changes in residential/commercial floorspace



Back

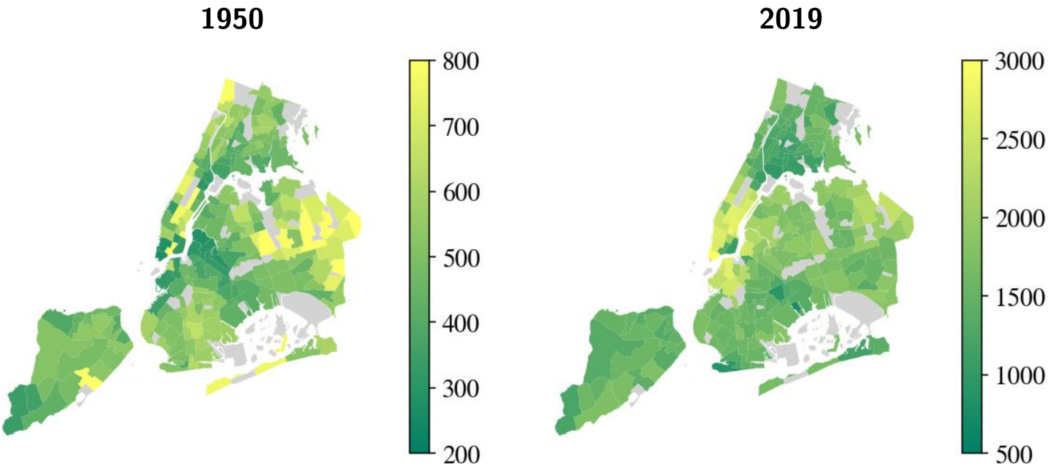
FAR allowances in NYC's zoning map





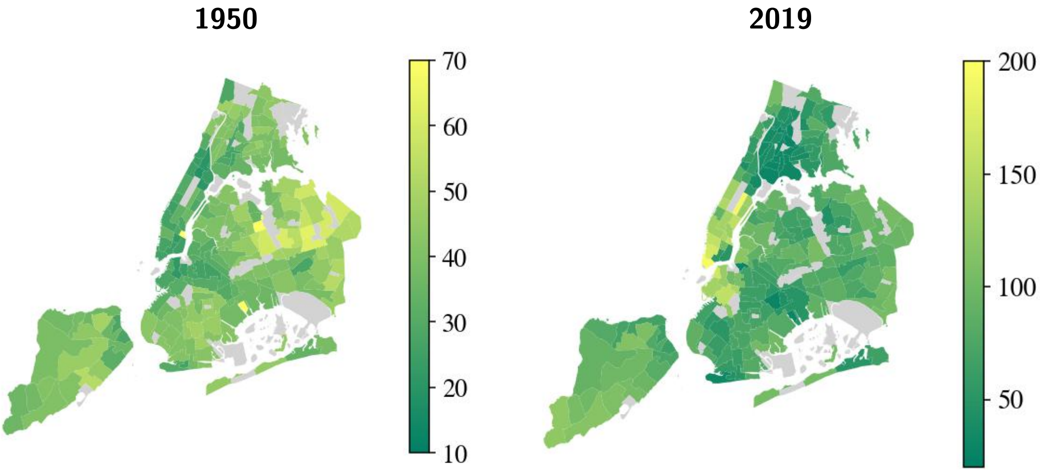
Back

Evolution of median residential rents ($/month)



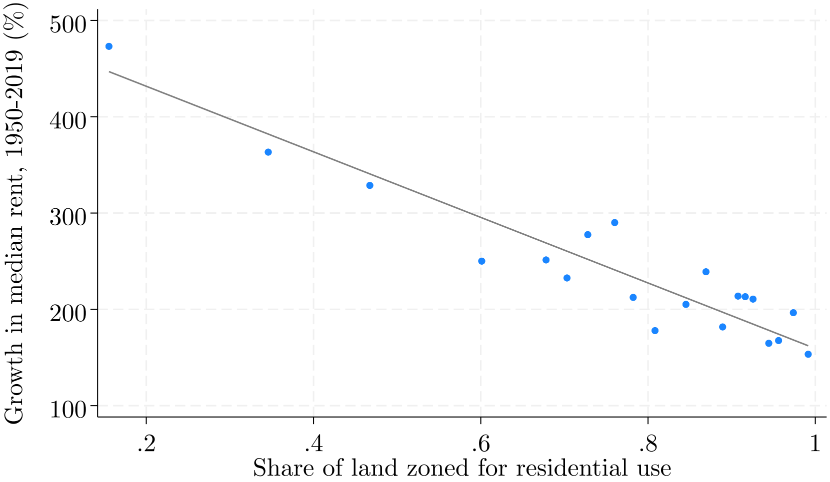
Back

Evolution of median household income ($k/year)



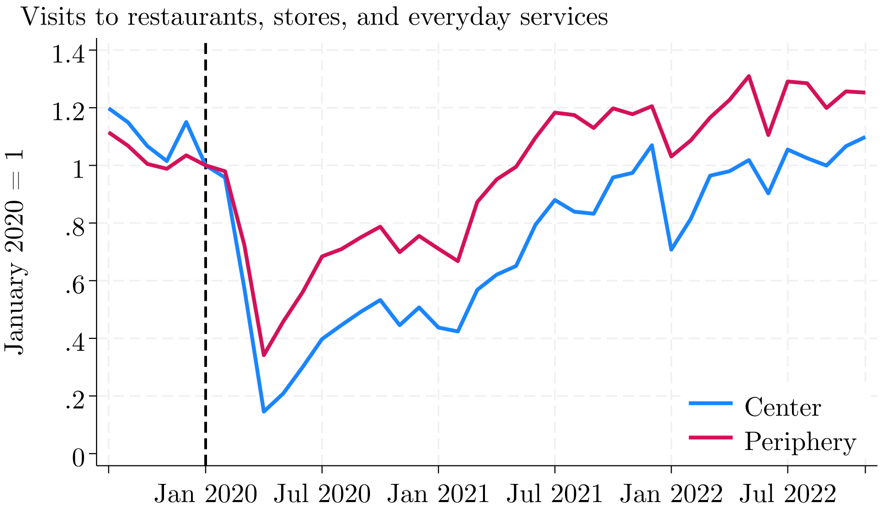
Back

Commercial-oriented areas became more attractive over time



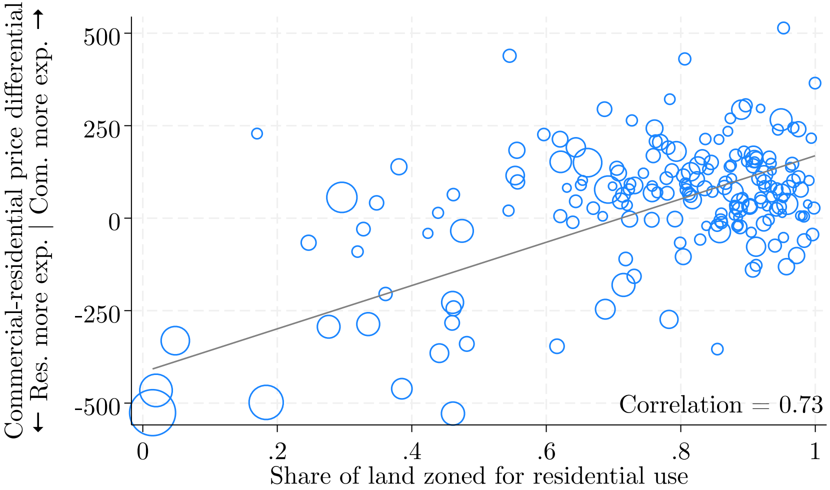
Back

Work from home will likely accelerate existing trends



Back

Use restrictions constrain the reallocation of land uses



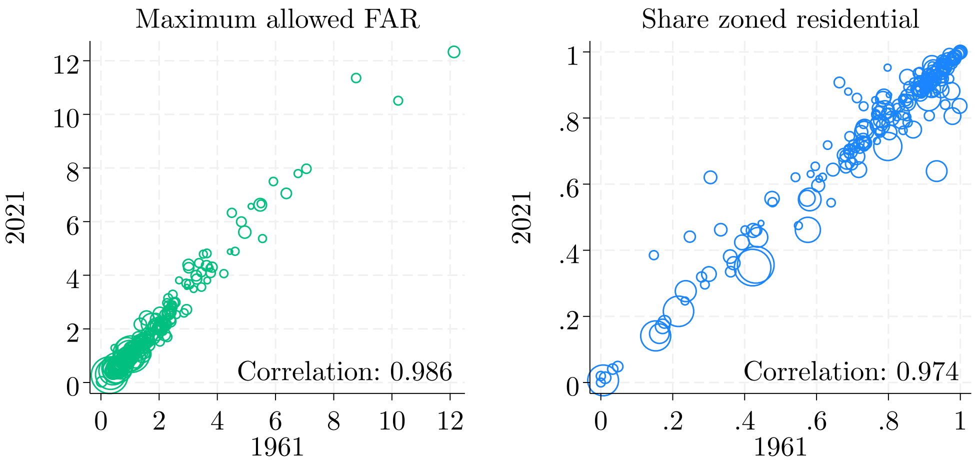
◂ Back

Zoning's goals

- Current zoning resolution adopted in 1961.
- Planners believed NYC has nearly reached its maximum size.
  - → Didn't view the zoning code as restrictive.
- Aims of the 1961 zoning resolution:
  1. Promote tower-in-the-park development.
  2. Better separate commercial/residential uses.
  - → De facto: stabilization of existing land uses.
- Current zoning regulations closely align with the 1961 ordinance.

Back

Since 1961: Persistence in zoning



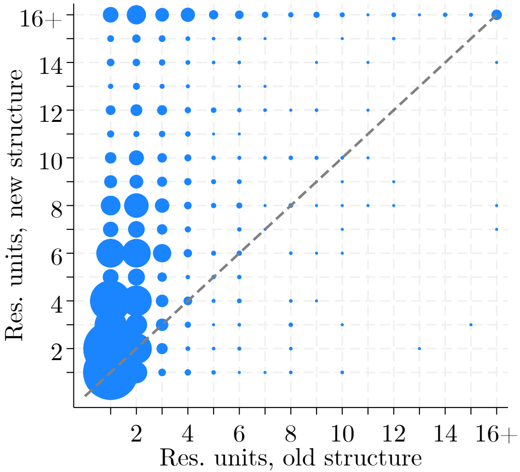
Back

Change in number of units

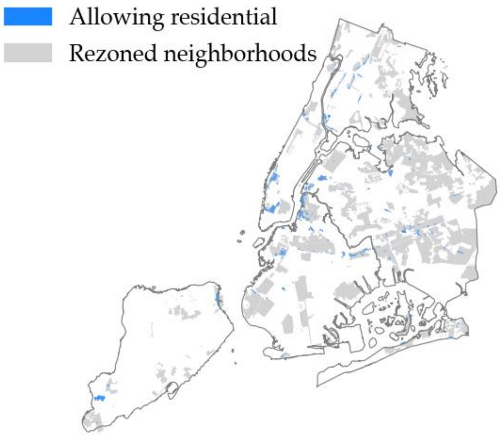
Residential-to-residential redevelopment

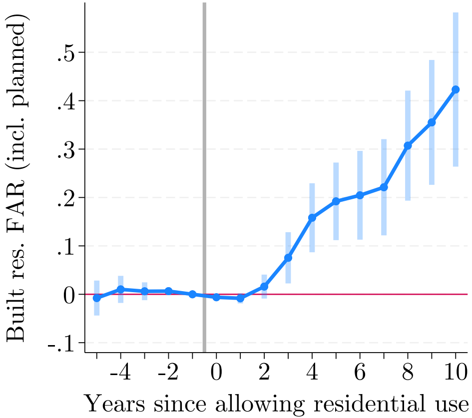
- New residential units are about 10% larger than the ones they replace.
- The number of units in new residential buildings is, on average, 3 times larger than in old structures.

Back



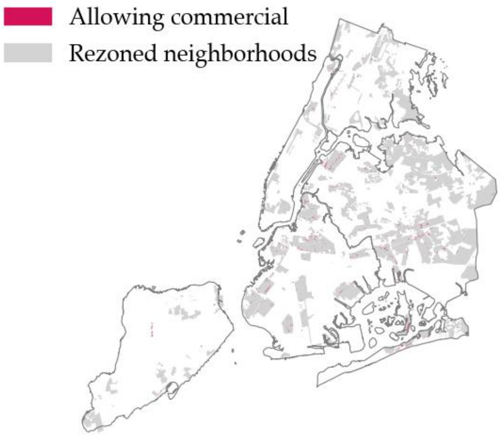
Effects of allowing residential use

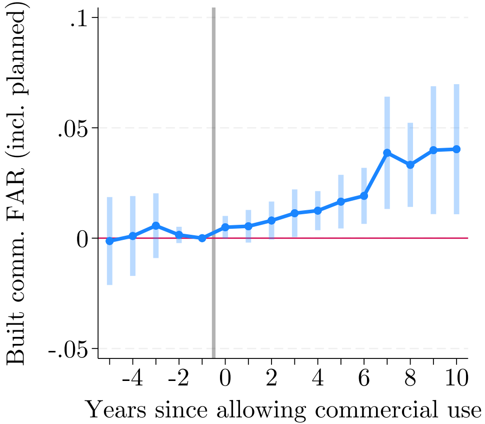




◂ Back (upzoning event study) ◂ Back (price differentials)

Effects of allowing commercial use





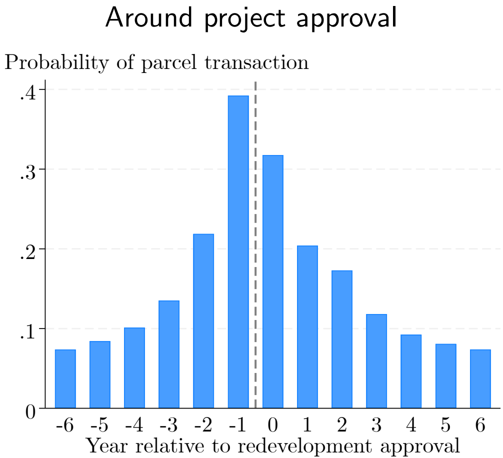
◂ Back (upzoning event study) ◂ Back (price differentials)

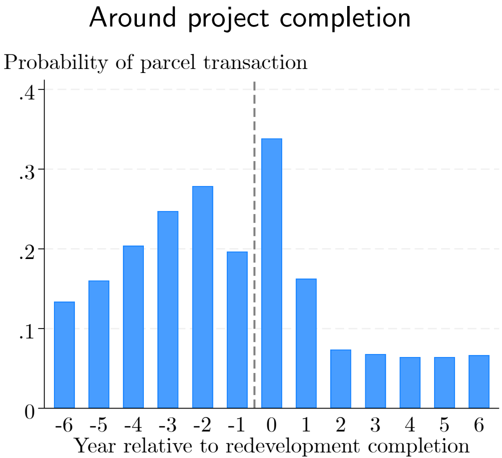
Probability of a parcel sale (probit regression)

	Probit coefficients	
Parcel sold		
Office space (% of total floorspace)	-0.076	(0.006)
Retail space (% of total floorspace)	-0.018	(0.004)
Garage space (% of total floorspace)	0.000	(0.007)
Storage space (% of total floorspace)	0.033	(0.009)
Factory space (% of total floorspace)	0.001	(0.007)
Hotel space (% of total floorspace)	-0.015	(0.020)
Other space (% of total floorspace)	-0.327	(0.006)
Condo/Coop	-0.684	(0.008)
Parcel in Bronx	0.051	(0.003)
Parcel in Brooklyn	0.038	(0.003)
Parcel in Queens	0.036	(0.003)
Parcel in Staten Island	0.023	(0.003)
Constant	-1.615	(0.003)
Observations	13,156,064	

Back

Parcel sales around redevelopment events





Back

Estimation details: Building values

- Using the sales data, I estimate separately for residential and commercial structures:

$$\log(\text{price}_s) = \underbrace{\rho^{\theta}_{1,n(s)}}_{\text{Neigh. FE}} + \underbrace{\rho^{\theta}_{2,bt(s)}}_{\text{Borough} \times \text{year FE}} + \boldsymbol{\beta} \underbrace{(\mathbf{x}_s - \bar{\mathbf{x}}_s)}_{\text{Structure characteristics}} + \nu_s$$

- Expected value of floorspace in a new building: $\bar{p}_{nt} = \exp(\hat{\rho}_{1,n} + \hat{\rho}_{2,b(n)t} + \hat{\sigma}_{\nu}/2)$
- Negative coefficient on (log) FAR, more negative for commercial.
  - → Less usable floorspace in tall buildings (e.g., because of mechanical space).
  - → Lower floor of commercial buildings is more valuable.

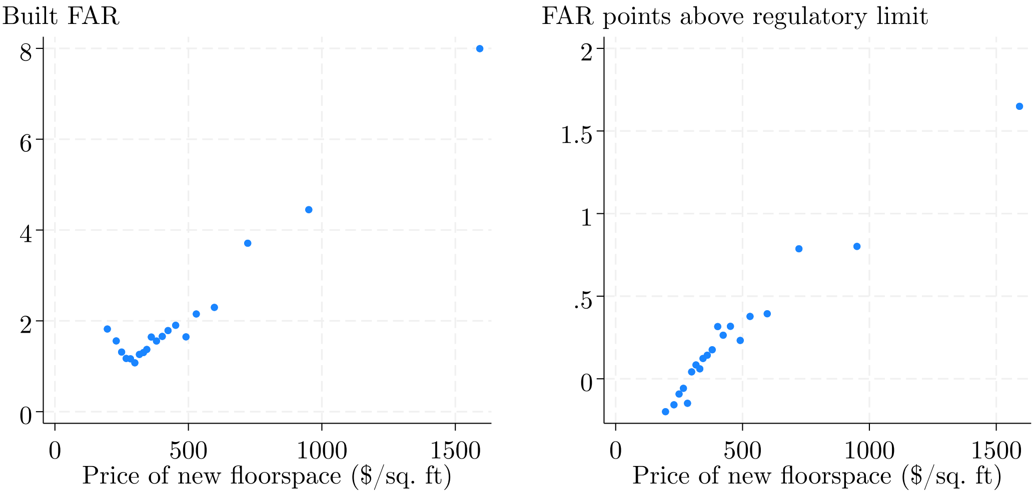
◂ Back

Hedonic regression coefficients

Back

	(1) Residential		(2) Commercial	
(log) Built FAR	-0.027	(0.002)	-0.091	(0.011)
(log) Unit size	-0.065	(0.002)	-0.108	(0.005)
Age	-0.002	(0.000)	0.000	(0.000)
Rent-stabilized	-0.336	(0.002)		
Landmark	-0.078	(0.006)	0.087	(0.060)
Grade A	0.162	(0.003)	0.222	(0.032)
Grade B	0.025	(0.002)	0.095	(0.019)
Grade C	0.007	(0.002)	-0.024	(0.019)
Brick	-0.155	(0.003)	-0.001	(0.084)
Frame	-0.068	(0.003)	-0.224	(0.118)
Masonry	-0.156	(0.003)	-0.071	(0.019)
Office building			0.147	(0.025)
Retail building			0.222	(0.021)
Garage building			0.004	(0.025)
Industrial building			0.030	(0.025)
Hotel			0.352	(0.048)
Neighborhood FE	Yes		Yes	
Borough $\times$ Year FE	Yes		Yes	
Observations	428,338		15,795	

Developers build taller when facing high prices



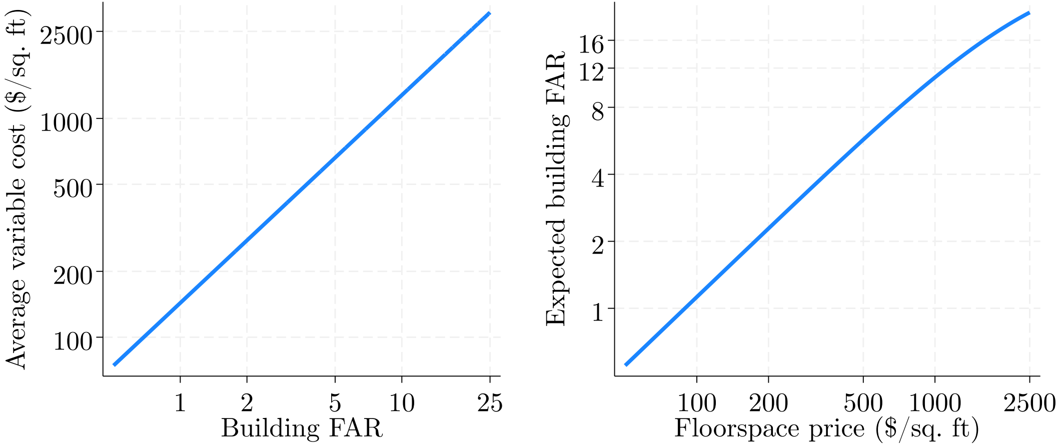
Back

Estimated cost parameters

(a) Variable cost parameters			
α^0	Baseline cost of materials	80.4	(2.3)
ζ	Capital cost share	0.51	(0.005)
σ_η	Cost shock standard deviation	1.08	(0.02)
(b) Fixed cost parameters			
δ^0	Base fixed cost	175.2	(7.7)
$\delta^{\text{demolition}}$	Demolition multiplier	1853.8	(37.6)
δ^{density}	Neighborhood density multiplier	318.5	(11.3)
$\delta^{\text{protected}}$	Protected parcels multiplier	0.77	(0.03)
(c) Profit shock parameters			
σ_ϵ^0	Base profit shock variance	73.7	(2.9)
$\sigma_\epsilon^{\text{demolition}}$	Demolition multiplier	138.8	(5.1)
$\sigma_\epsilon^{\text{density}}$	Neighborhood density multiplier	248.2	(4.9)

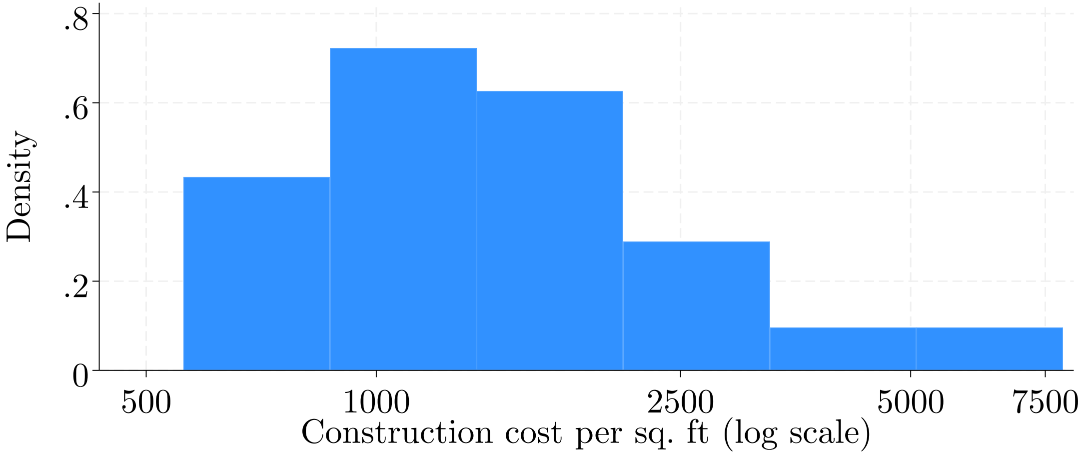
Back (variable costs) Back (fixed costs)

Construction costs



Back

Construction costs for skyscrapers



Back

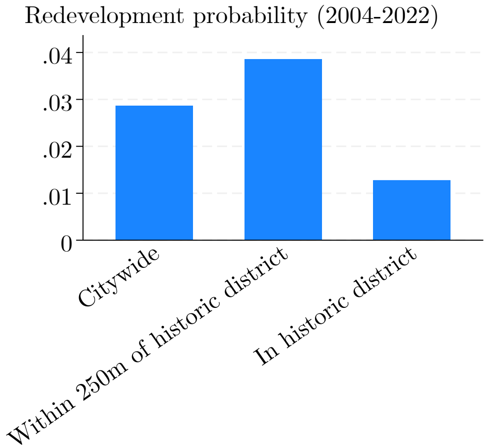
Fixed costs parameterization

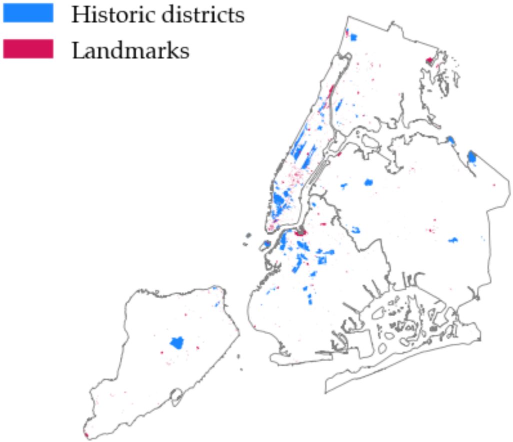
$$FC_{it} = (\delta^0 + \delta^{\text{demolition}} h_{it}^{\text{old}} + \delta^{\text{density}} \bar{h}_{n(i)t})(1 + \delta^{\text{protected}} \mathbf{1}_{it}^{\text{protected}}).$$

- δ^0: Baseline fixed cost.
- h_{it}^{old}: FAR of building to demolish.
- $\bar{h}_{n(i)t}$: Average neighborhood density.
- $\mathbf{1}_{it}^{\text{protected}}$: Landmarked parcel or in historic district.

Back

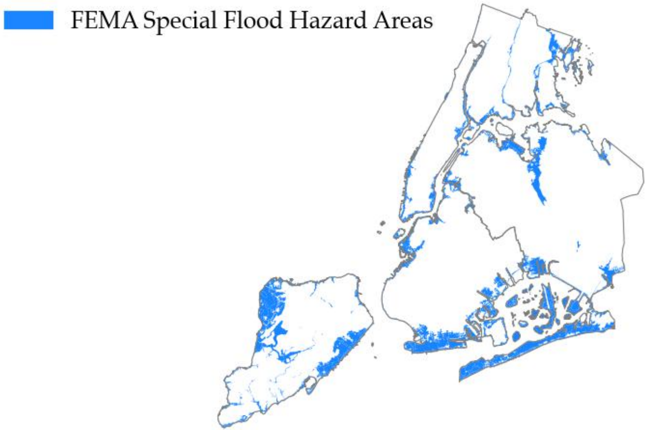
Protected areas





◂ Back (fixed costs) ◂ Back (counterfactuals)

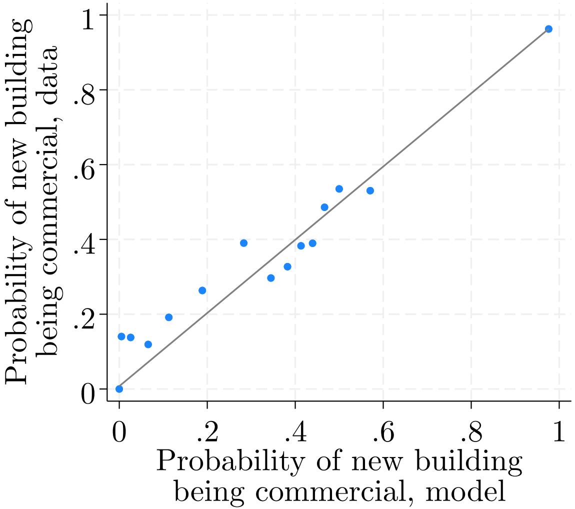
Flood zones



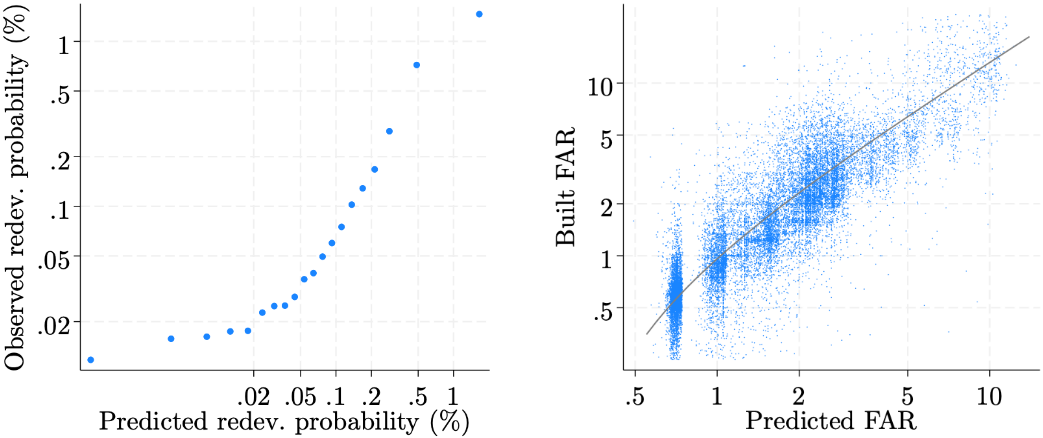
◂ Back

Model fit: Residential vs. commercial construction

Back



Fit on extensive and intensive margins



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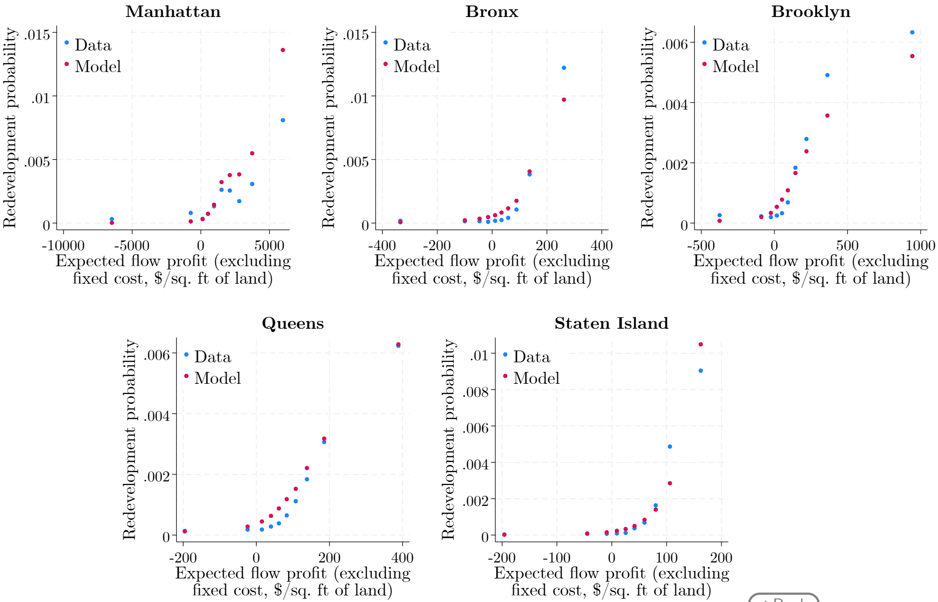

Manhattan
Bronx
Brooklyn
Queens
Staten Island
Data
Model
Redevelopment probability
Expected flow profit (excluding fixed cost, $/sq. ft of land)
0
.005
.01
.015
-10000
-5000
5000
-400
-200
200
400
.002
.004
.006
-500
500
1000
.008
-100
100

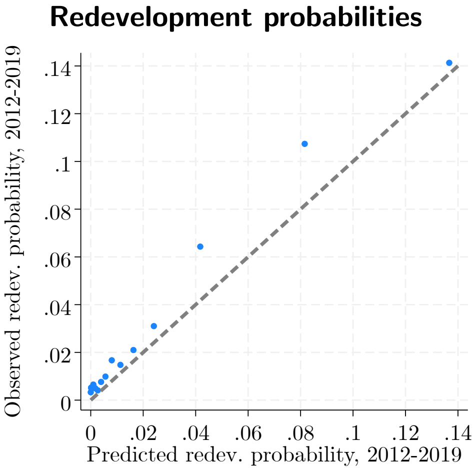
◂ Back

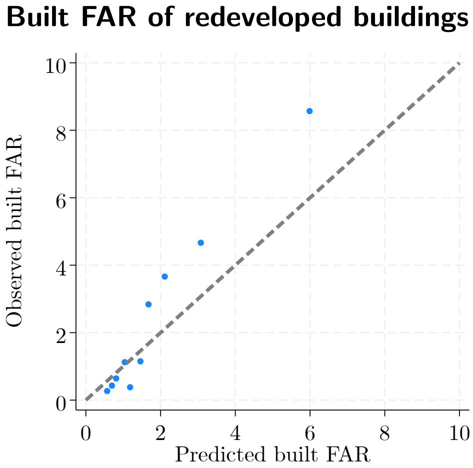
Out-of-sample model fit ◂ Back

- Re-estimate the model using first half of the data (2004–2011):
  - → Price levels;
  - → Preferences and location fundamentals;
  - → Sale probabilities;
  - → Variable costs;
  - → Fixed costs.
- Predict the evolution of the city over 2012–2019.

Out-of-sample model fit: Parcel level

Back



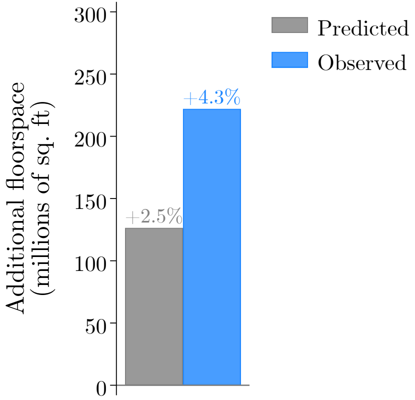


→ Differences explained by high price growth post-2012.

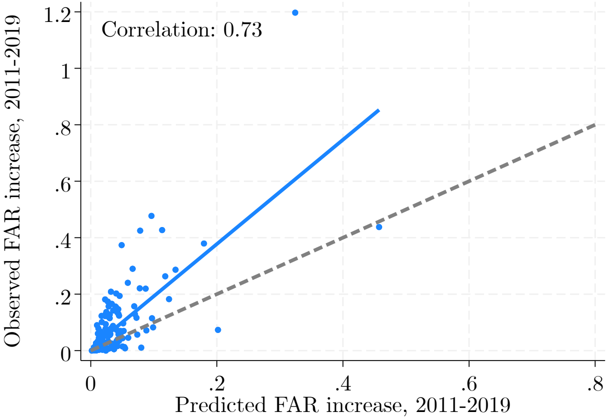
Out-of-sample model fit: Aggregate growth

Back

Citywide floorspace growth



Neighborhood-level growth



→ Differences explained by high price growth post-2012.

Demand model estimation (1)

- Calibrate β to 0.1 using the ACS.
- Estimate the shape of z^W at 4.4 using the commuting data.
- Calibrate wages from the number of people working in each location.
  $\rightarrow$ High-wage locations attract more workers, and from further away.
- Calibrate amenities $\boldsymbol{B}$ with residential prices and the number of residents in each location.
  $\rightarrow$ Locations attracting many residents despite high prices must have high amenities.
- Calibrate subsistence levels $\bar{\boldsymbol{h}}$ with the total housing consumption in each location.
  $\rightarrow$ Higher levels of $\bar{h}$ lead to more housing consumption.
  $\rightarrow$ Average $\bar{h}$ of 224 sq. ft (IQR = [170 sq. ft, 265 sq. ft])

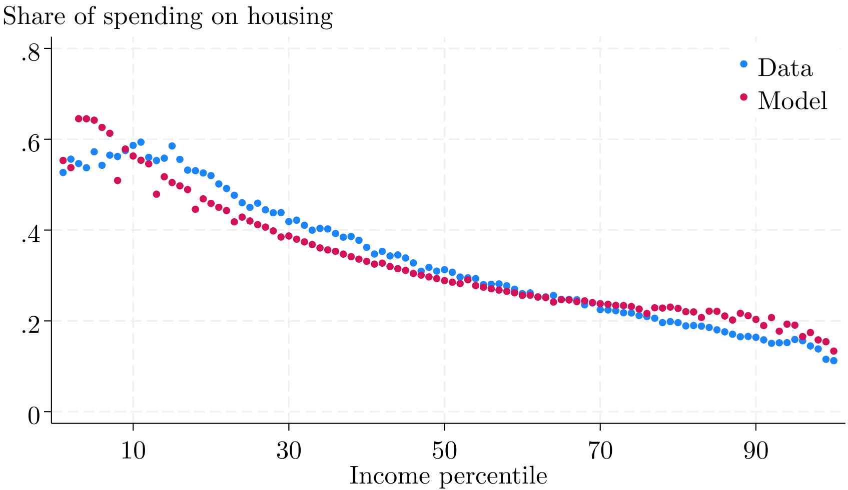
Demand model estimation (2)

- Calibrate the shape of z^H to 2.9 match the variance of average neighborhood incomes.
  → Higher variance of $z^H \implies$ less sorting across neighborhood by income.

- Calibrate productivities $\boldsymbol{A}$ and floorspace shares in production $\boldsymbol{\alpha}$ using data on commercial floorspace quantities and prices, the number of jobs in each location, and the calibrated wages.
  → Productivity is estimated to be higher near the center of the city.
  → α averages 0.18 across neighborhoods (IQR $=$ [0.14, 0.21]). Close to benchmarks in the literature (0.2 in Ahlfeldt et al., 2015; 0.16 in Greenwood, Hercowitz, and Krusell, 1997).

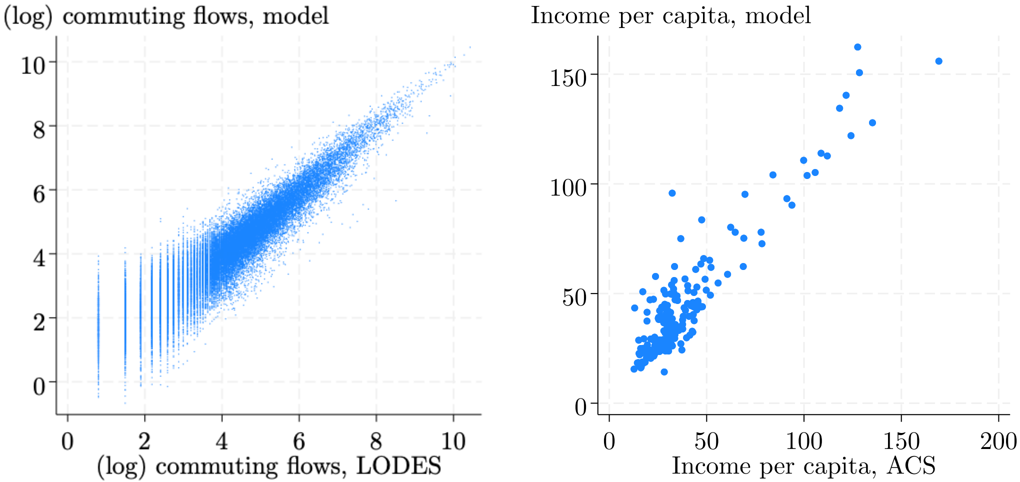
Back

Engel curve for housing



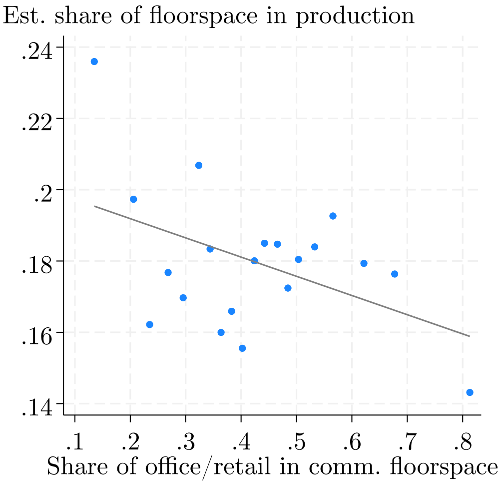
Back

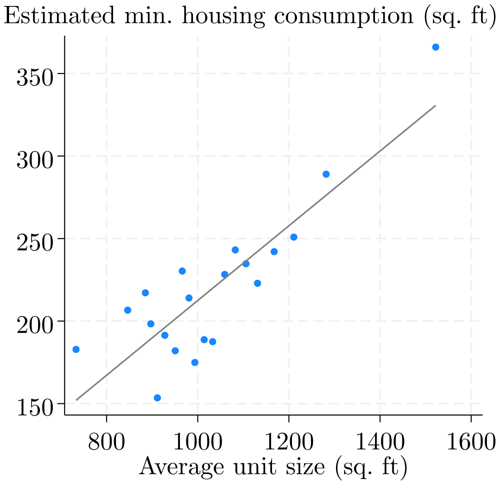
Demand model fit: Commuting flows and sorting



◂ Back

Demand model fit: α and $\bar{h}$





Back

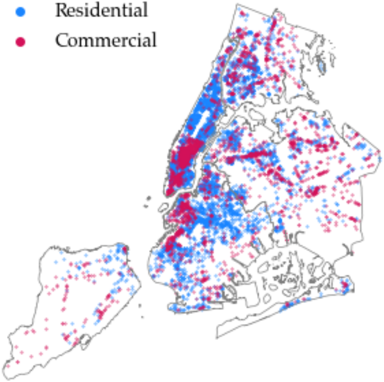
Amenity	**Correlation between share of satisfied residents and** B
Neighborhood cleanliness	0.47
Control of street noise	0.07
Household garbage pick-up	0.33
Recycling services	0.34
Snow removal	0.77
Rat control	0.36
Bike safety	0.25
Pedestrian safety	0.53
Street maintenance	0.53
Parking enforcement	0.74
Storm water drainage and sewer maintenance	0.57
Availability of healthcare services	0.52
Availability of cultural activities	0.60
Neighborhood parks	0.86
Fire protection services	0.75
Emergency medical services	0.73
Neighborhood public safety	0.66
Bus services	0.66
Subway services	0.71
Public services	0.63

Back

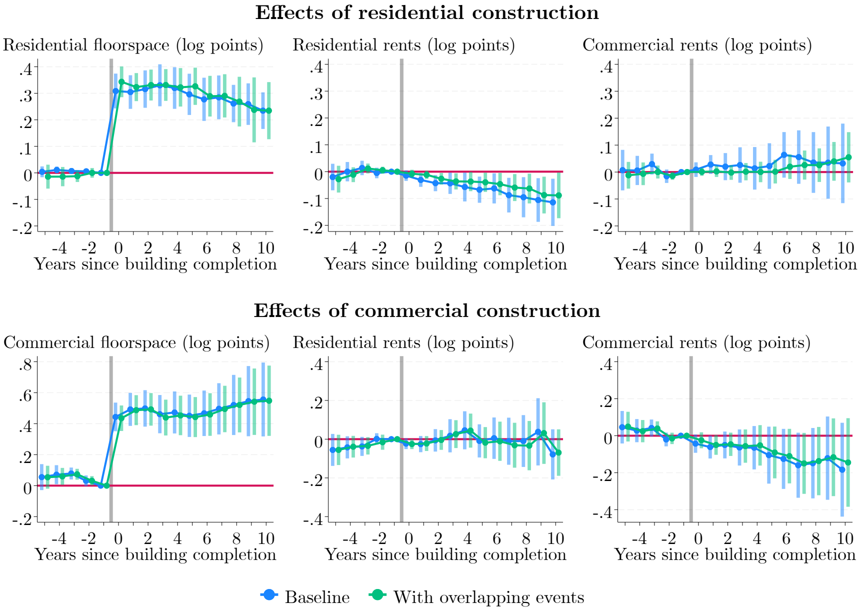
Large construction events

Construction events

Buildings with rent data

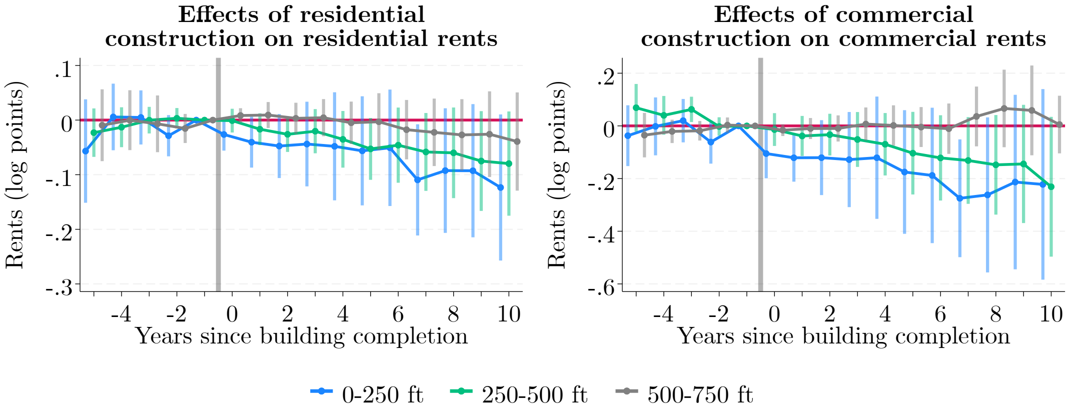


Back


Effects of residential construction
Residential floorspace (log points)
Residential rents (log points)
Commercial rents (log points)
.4
.3
.2
.1
0
-.1
-.2
-4 -2 0 2 4 6 8 10
Years since building completion
Effects of commercial construction
Commercial floorspace (log points)
Residential rents (log points)
Commercial rents (log points)
.8
.6
.4
.2
0
-.2
-.4
Years since building completion
Baseline
With overlapping events

Back

Spatial decay of price effects

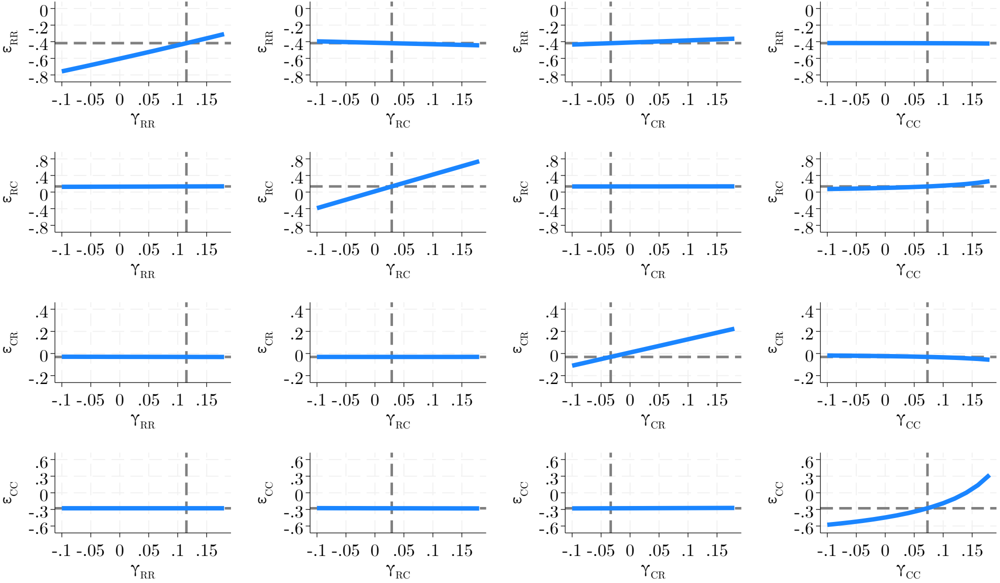


Back

Estimated spillover parameters

Parameter	Interpretation	Calibrated value	Targeted elasticity
γ^{RR}	Effect of residents on amenities	0.11	$\varepsilon^{RR} =$ -0.42
γ^{RC}	Effect of residents on productivity	0.03	$\varepsilon^{RC} =$ 0.14
γ^{CR}	Effect of jobs on amenities	-0.03	$\varepsilon^{CR} =$ -0.03
γ^{CC}	Effect of jobs on productivity	0.07	$\varepsilon^{CC} =$ -0.28

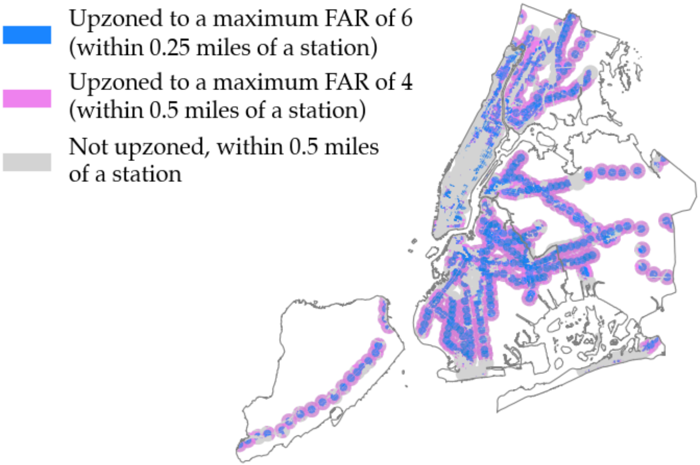
Back



ε_RR
ε_RC
ε_CR
ε_CC
ϒ_RR
ϒ_RC
ϒ_CR
ϒ_CC
0
-.2
-.4
-.6
-.8
.8
.4
0
-.4
-.8
.4
.2
0
-.2
.6
.3
0
-.3
-.6
-.1 -.05 0 .05 .1 .15

Back

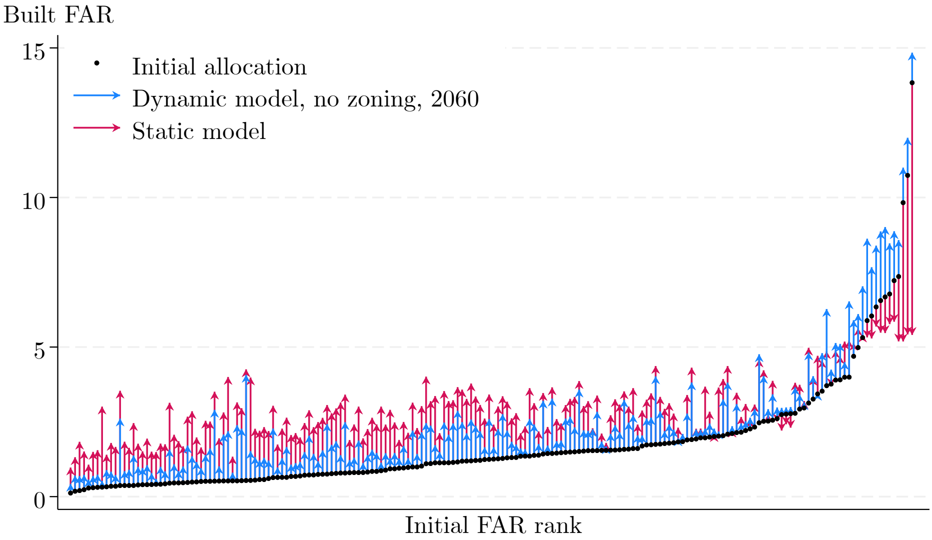
Transit-Oriented Development



→ Upzones 37% of the city, mostly in outer boroughs.

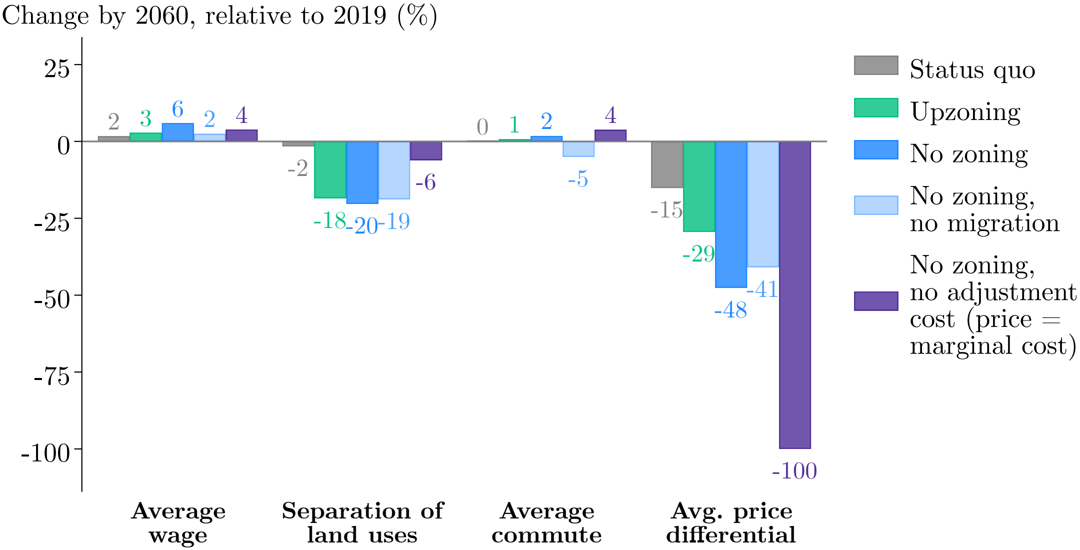
Back

Dynamic vs. static model



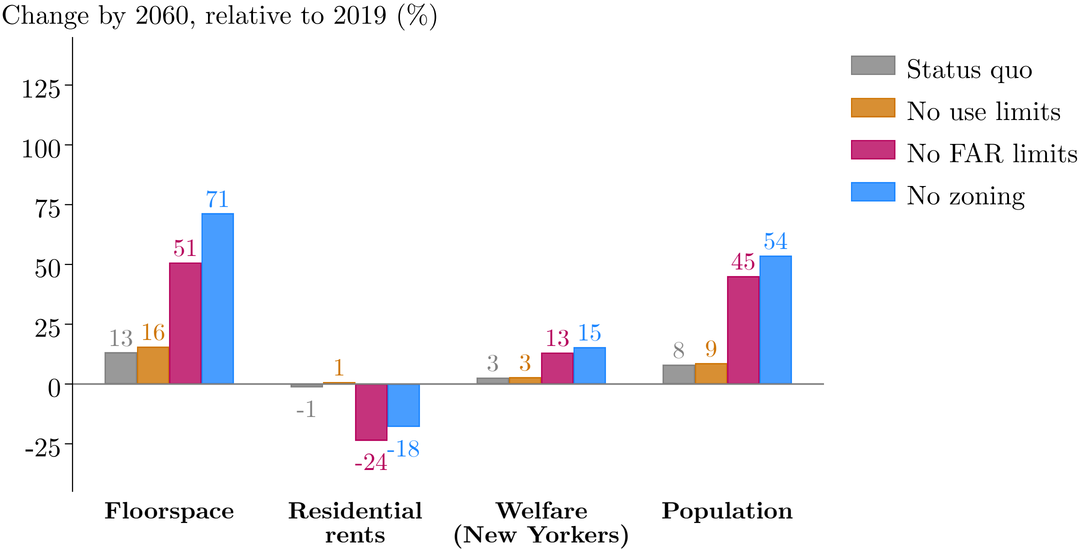
Back

Additional outcomes



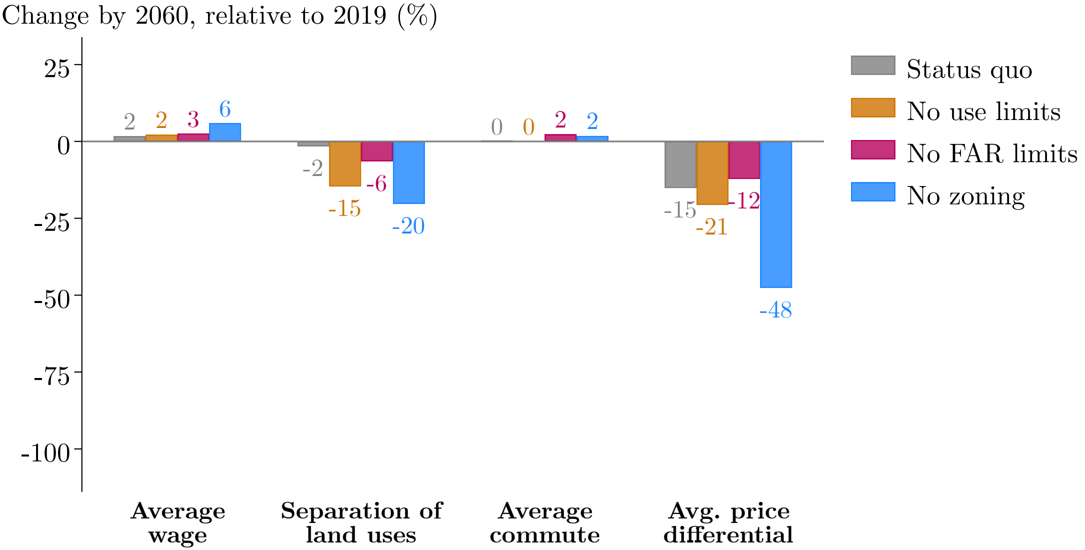
Back

Use vs. FAR limits



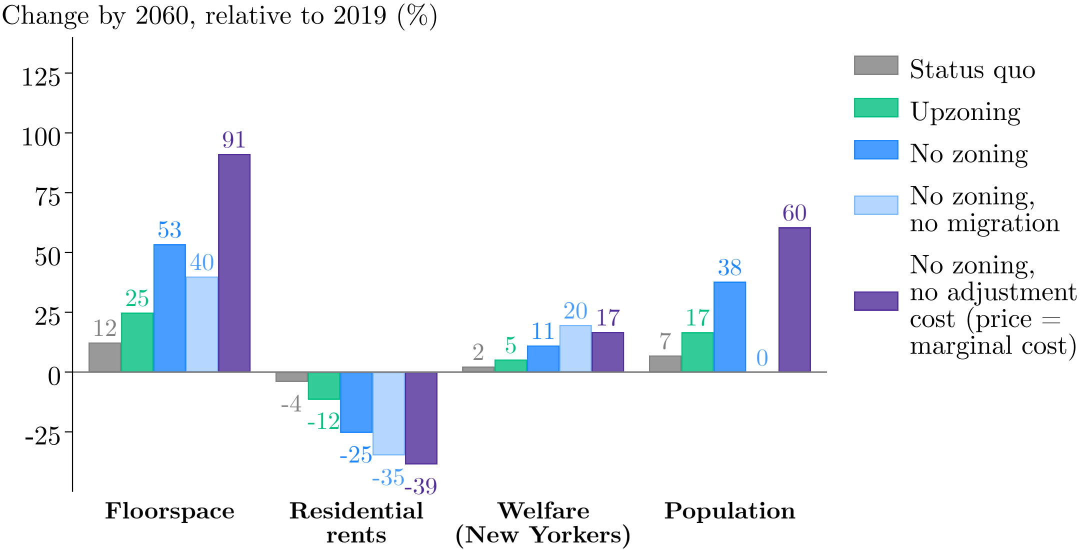
Back

Use vs. FAR limits



Back

No spillovers



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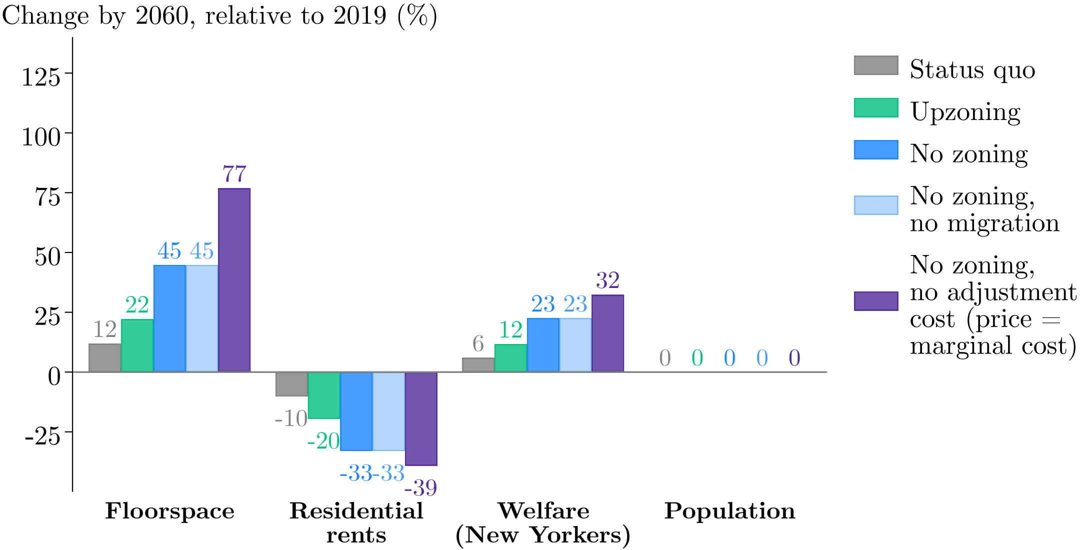
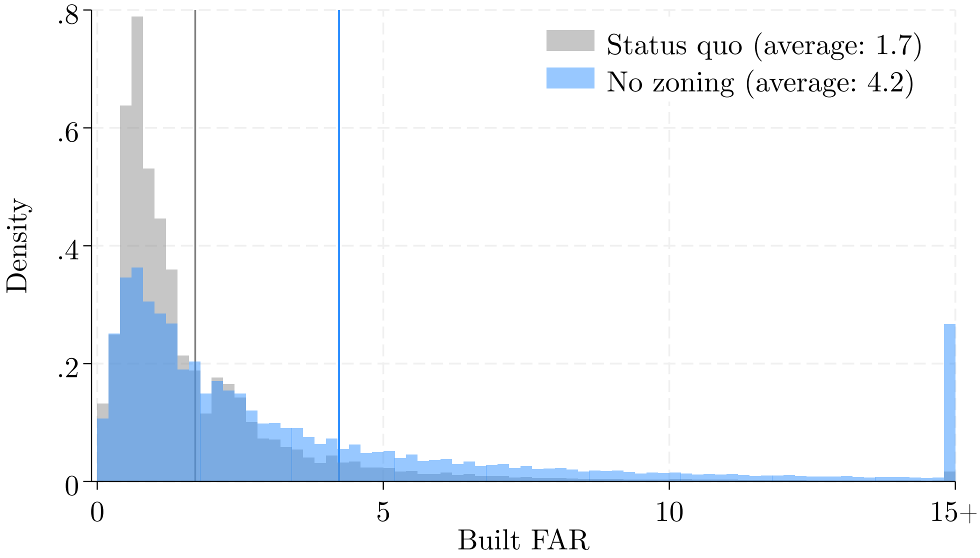
No migration



Back

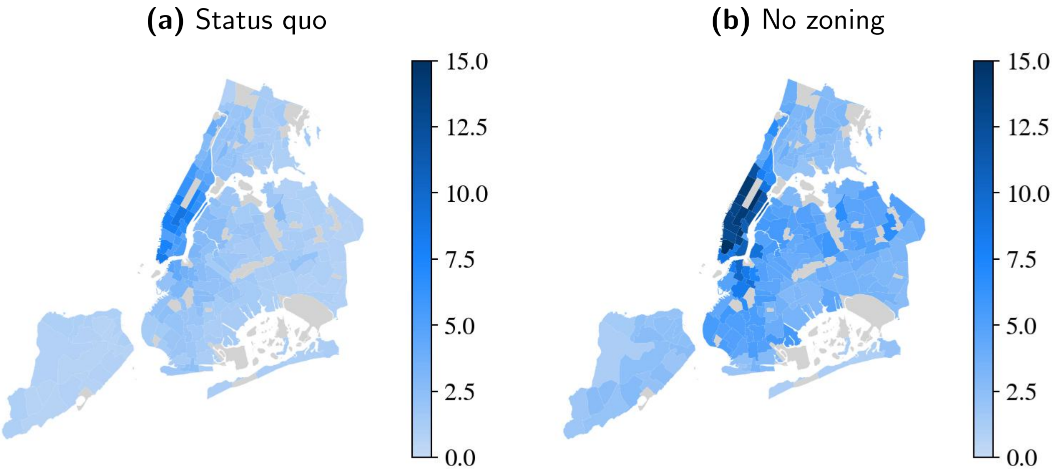
FAR distribution of new buildings

Back



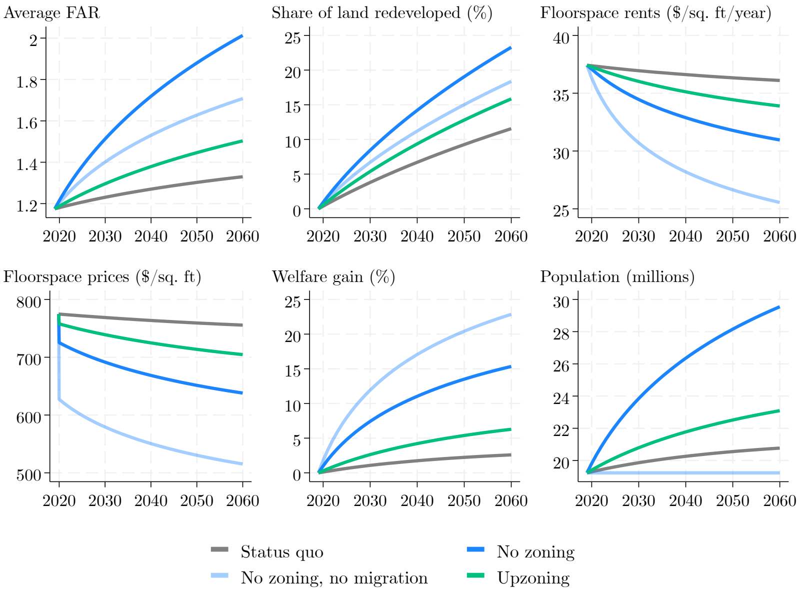
Average FAR of new buildings

◂ Back



Transition paths

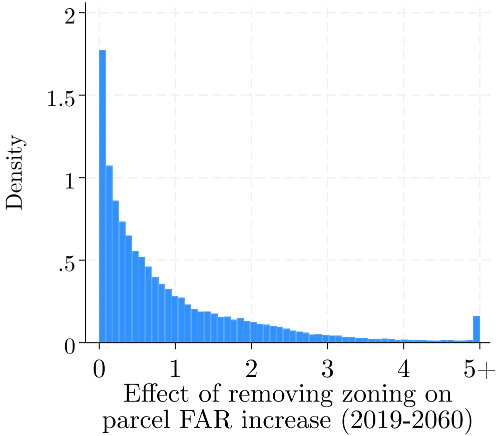
Back



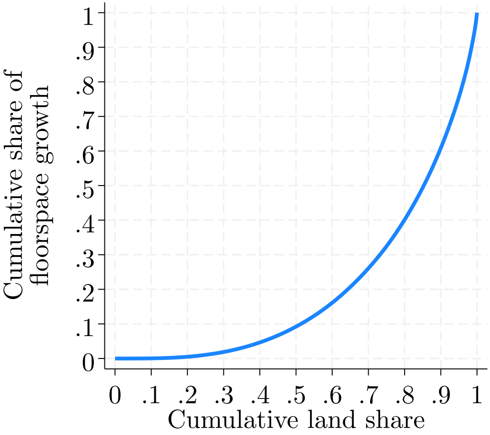
Upzonings' effects are concentrated

Back

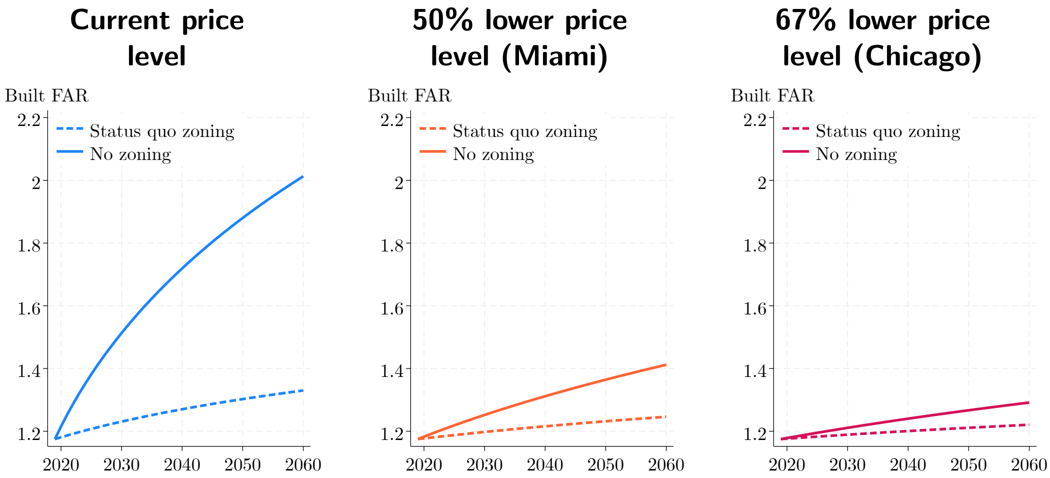
Parcel-level distribution of FAR increases



Spatial concentration of floorspace growth



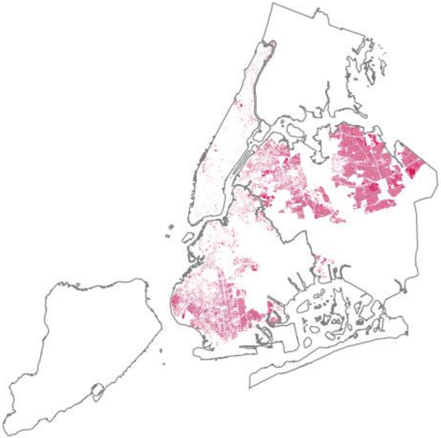
Effects of the aggregate price level



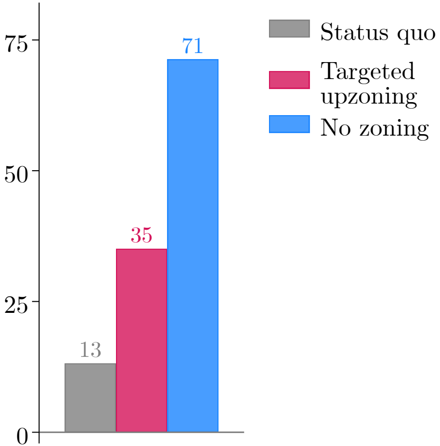
Back

Targeted upzoning

Upzoned parcels



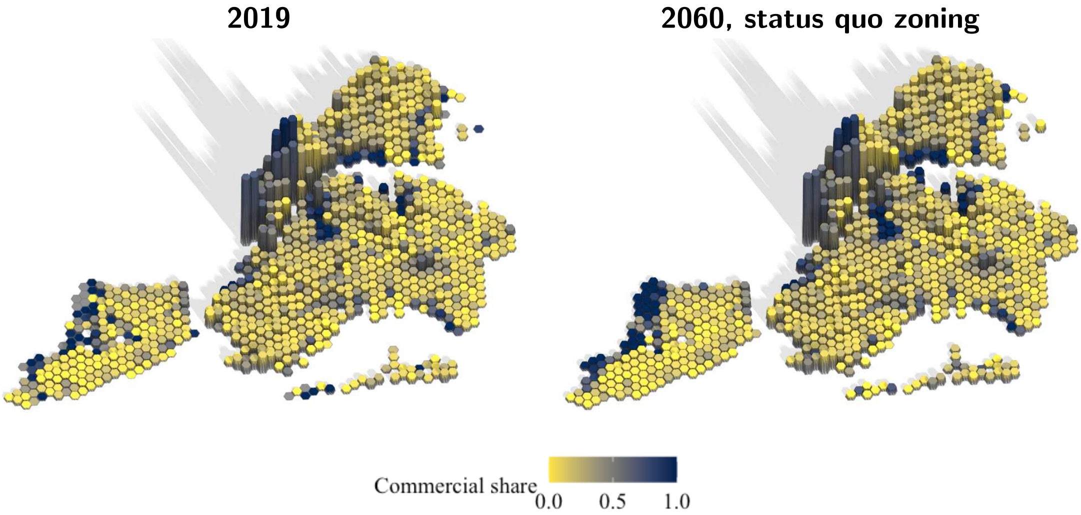
Floorspace growth, 2019–2060



Back

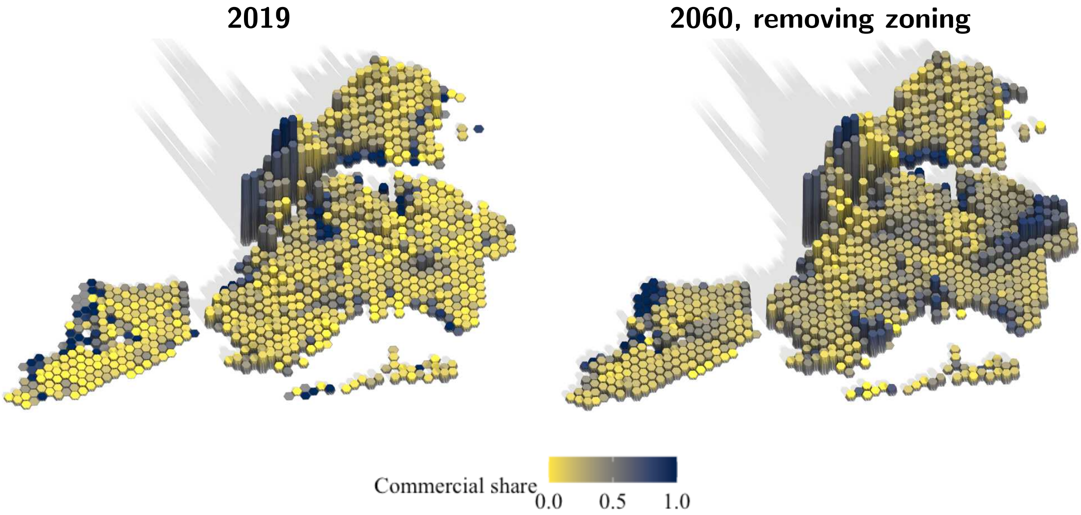
Effects on city structure

Back

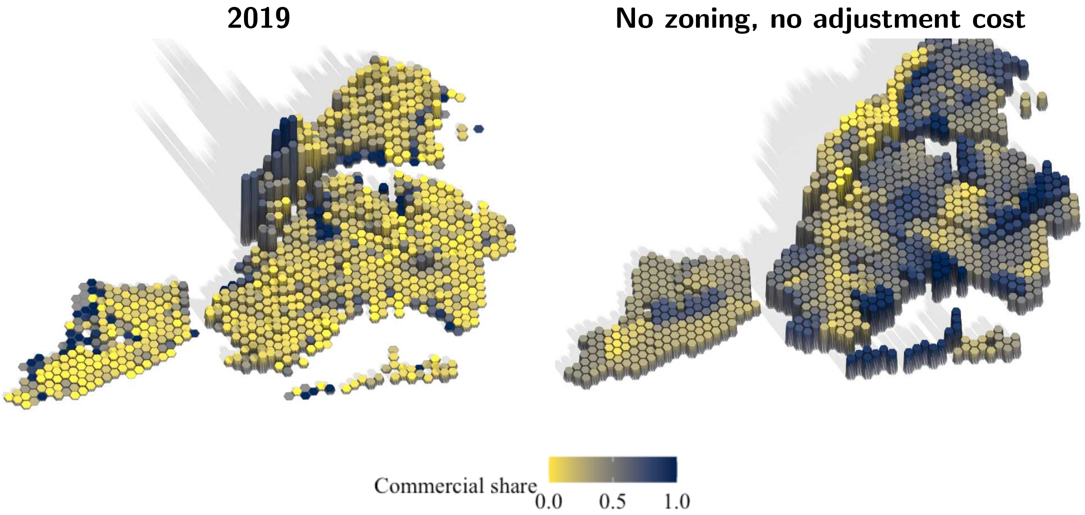


Effects on city structure

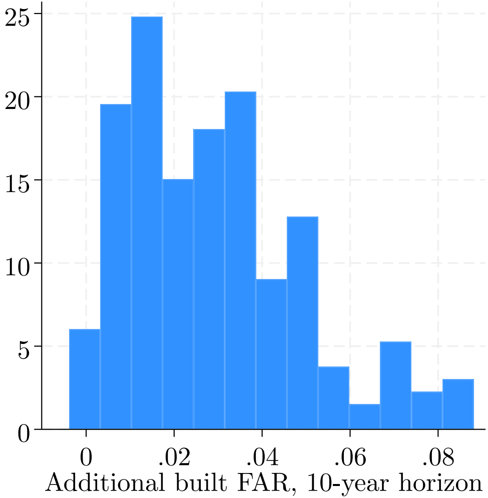
Back



Effects on city structure ◂ Back



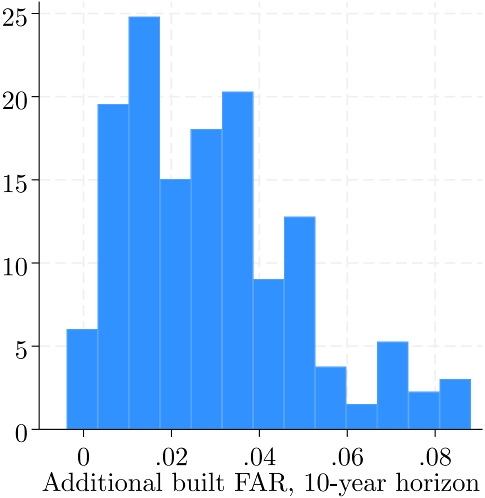
Effects of upzoning vary widely across neighborhoods



- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
  → For comparison with event study.

Back

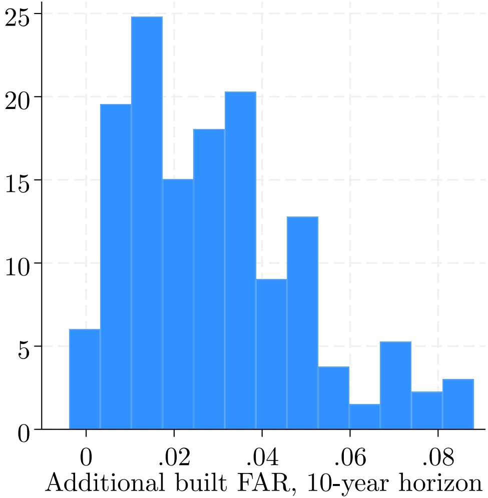
Effects of upzoning vary widely across neighborhoods



- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
  → For comparison with event study.
- Wide heterogeneity in effects.
  → Important consequences if upzoning is politically costly.

Back

Effects of upzoning vary widely across neighborhoods

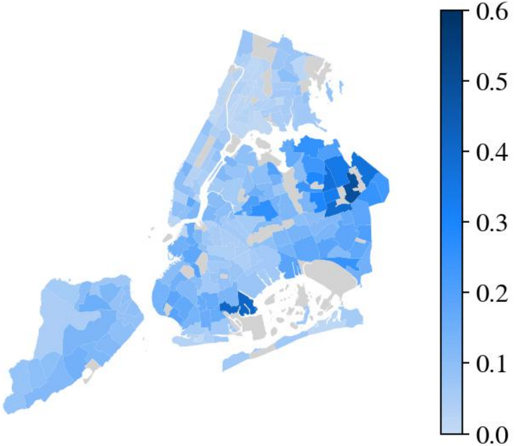


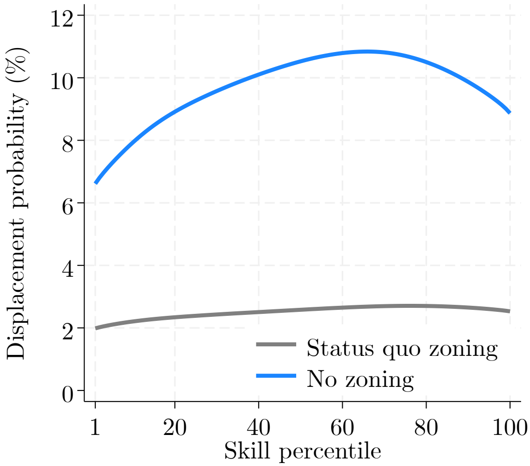
- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
  - → For comparison with event study.
- Wide heterogeneity in effects.
  - → Important consequences if upzoning is politically costly.
- Realized upzonings were in areas ripe for redevelopment.
  - → They were "selected on gains."

Back

Richer households are more exposed to redevelopment

Share of land redeveloped by 2060
(no zoning regulations)

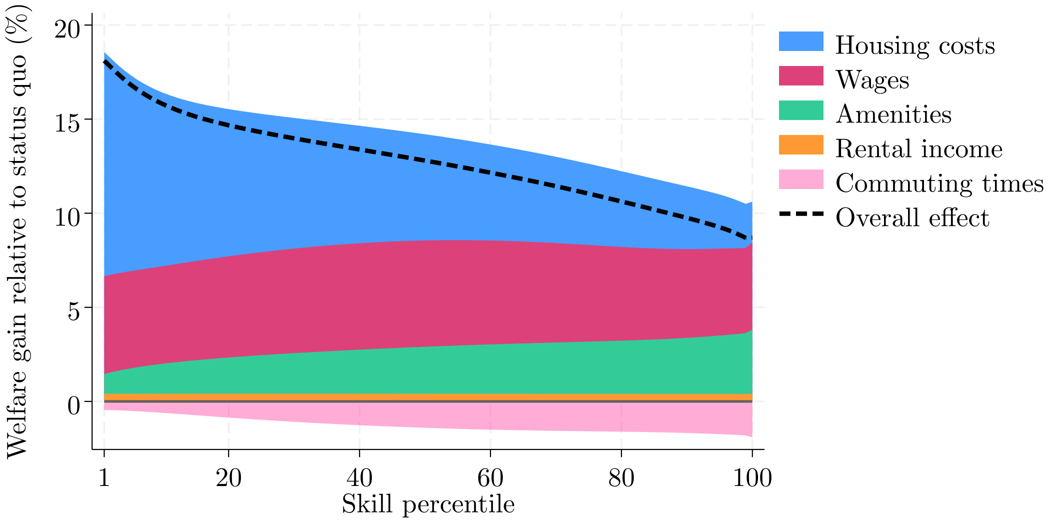




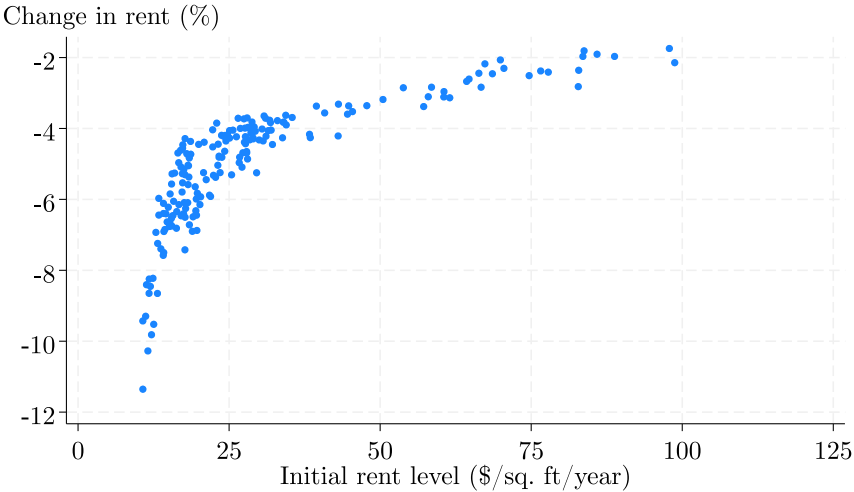
Back

Decomposition of the welfare gains from removing zoning

Back

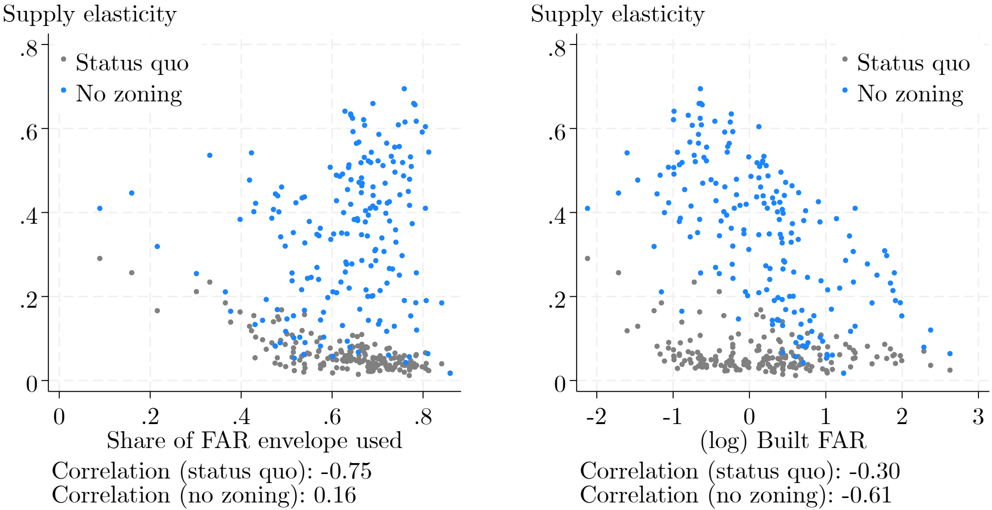


Effects of a 10% increase in the city's floorspace on rents



Back

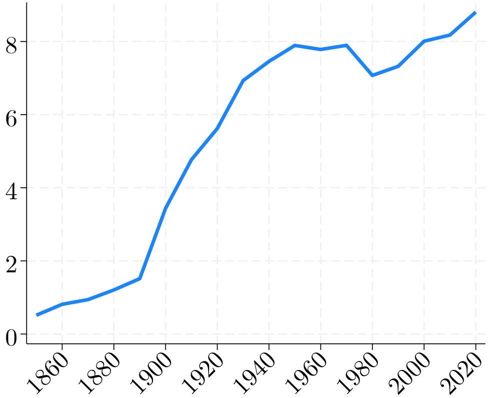
Determinants of supply elasticities



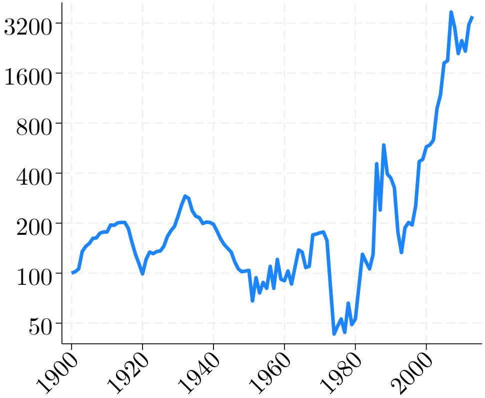
Back

Historical population and land values

(a) NYC population (millions)

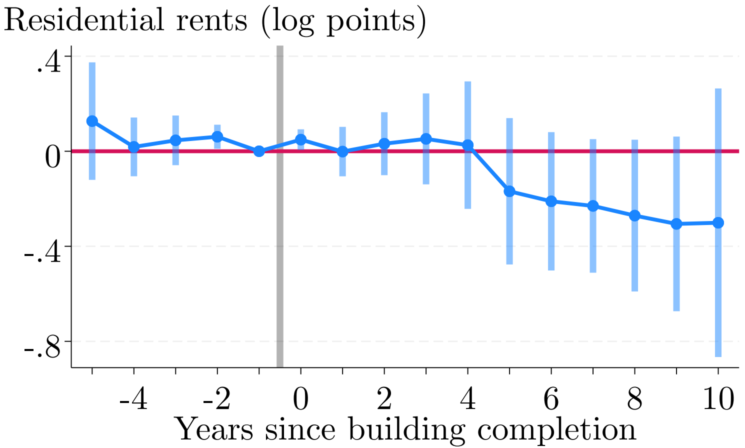


(b) Manhattan land values (1900 = 100)



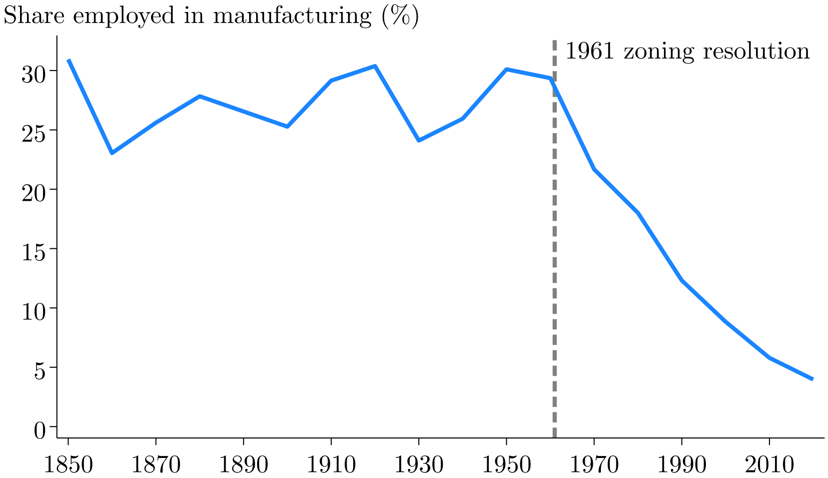
Back

Effects of industrial construction on residential rents



◂ Back (Spillovers) ◂ Back (Why zoning)

Manufacturing in NYC has collapsed since 1961



Back